FIRST EDITION · 2026

Python
A Complete Course

From the silicon up: a textbook for those who want to know not just how to write Python, but why Python is the way it is.

P

Volume I · Foundations & Core Language

With Numbered Examples, Exercises, and Complete Solutions

Preface

There are, by recent count, more than three hundred books on the Python programming language in print. This one differs from most of them in two respects.

The first is that it does not begin with Python. It begins, instead, with the machine. For three chapters you will read no Python at all. You will read about transistors, bits, instructions, processes, and the long history of how human beings learnt to make machines obey them. This is not preamble. It is the foundation, without which every later chapter would be a story told from the middle.

The second is that it does not assume you wish to be told. It assumes you wish to *understand*. Where another book might say "use a list", this book will tell you what a list is in memory, why appending to it is fast and prepending is not, when to choose a tuple instead, and what CPython's implementers did to make the choice matter as little as possible. The aim is fluency, not just fluency in writing Python but fluency in *thinking* about it.

How to Use This Book

Read each chapter in order. The examples are numbered (Example 4.3 is the third example in Chapter 4); the exercises follow each chapter; the complete solutions appear at the end of every chapter. Do the exercises before reading the solutions, even when you are certain of the answer. Programming is a motor skill; the muscle memory of typing `def` and `:` at the right moment is built only by typing them.

On Depth and Further Reading

Every topic in this book is the subject of its own books, papers, and lifelong careers. We cannot, and will not pretend to, cover everything. Where a subject deserves a book of its own, you will find a "Further Reading" box at the end of the section, pointing to the resources we judge to be the best on the internet. This book is a foundation; those resources are where you continue building.

Conventions

Cream code blocks are Python source. Grey blocks marked **Output** are what the interpreter prints. Yellow-bordered boxes contain Examples. Blue-bordered boxes contain Exercises. Indigo-bordered boxes contain links to Further Reading. Grey boxes contain Definitions of vocabulary you will use throughout the book.

Prerequisites

None, beyond curiosity and a willingness to read carefully. The mathematics will not exceed what you learnt in school. The first three chapters assume no programming background at all.

Contents

Volume I · Twenty-Two Chapters in Five Parts

PART I · THE GROUND

Foundations Beneath the Language

1. The Computer Beneath 1

2. The Lineage of Python 38

3. The Object Model 72

PART II · THE PRIMITIVES

Values and How They Behave

4. Numbers in Five Forms 108

5. Strings & Text 142

6. Booleans & Truthiness 186

7. None & the Sentinel 204

PART III · THE CONTAINERS

Structures for Collections

8. Lists in Depth 218

9. Tuples & Named Tuples 252

10. Dictionaries & Hash Tables 270

11. Sets & Frozensets 308

12. The collections Module 328

PART IV · FLOW & FUNCTION

The Shape of a Program

13. Control Flow 346

14. Iteration in Depth 368

15. Functions 394

16. Closures & Decorators 438

17. Comprehensions & Generators 462

PART V · LARGER PIECES

From One File to a Project

18. Modules, Packages, Environments 488

19. Classes & Objects 516

20. Files & the Outside World 562

21. Errors, Exceptions, Traces 590

22. A Workshop of Real Problems 618

CHAPTER 1 · PART I · THE GROUND

The Computer Beneath

Before you can program a machine, you should know what kind of machine it is. This chapter is a brief tour of the device, and of the five thousand years of effort that produced it, laying down only as much groundwork as the rest of the book will need.

> *A computer is a device which obeys instructions. We have built them so well that we now have to remember that this is all they do.*
>
> — Maurice Wilkes, designer of EDSAC (1949), the first practical stored-program computer

The book you are holding teaches a programming language called Python. But Python is, in the end, a thin layer of human convenience laid over a very specific kind of machine, and that machine has properties (speed, capacity, behaviour) which constrain everything you will ever do with it. If you want to be a competent programmer, you can begin with the language and learn what is beneath only when you need to. If you want to be a thoughtful one, you begin with what is beneath. We are taking the second route.

This chapter is not a course in computer architecture. A proper course in computer architecture is, by itself, a year of university study; we point at the best ones in the Further Reading boxes throughout. What you will find here is the minimum that the rest of this book assumes: a vocabulary for the parts of a computer, an intuition for their relative speeds, and a sense of the long human history that led to the device on your desk.

1.0 Before Electronics: Five Thousand Years of Calculating Machines

It is tempting, sitting in front of a glowing rectangle, to imagine that computing began with the silicon chip. It did not. The desire to delegate calculation to a machine is older than writing, and the lineage of devices that have answered that desire reaches back five thousand years. Knowing this lineage will not, by itself, make you a better Python programmer; but it will make you a less arrogant one, and that is its own kind of skill.

The Abacus, circa 2700 BCE

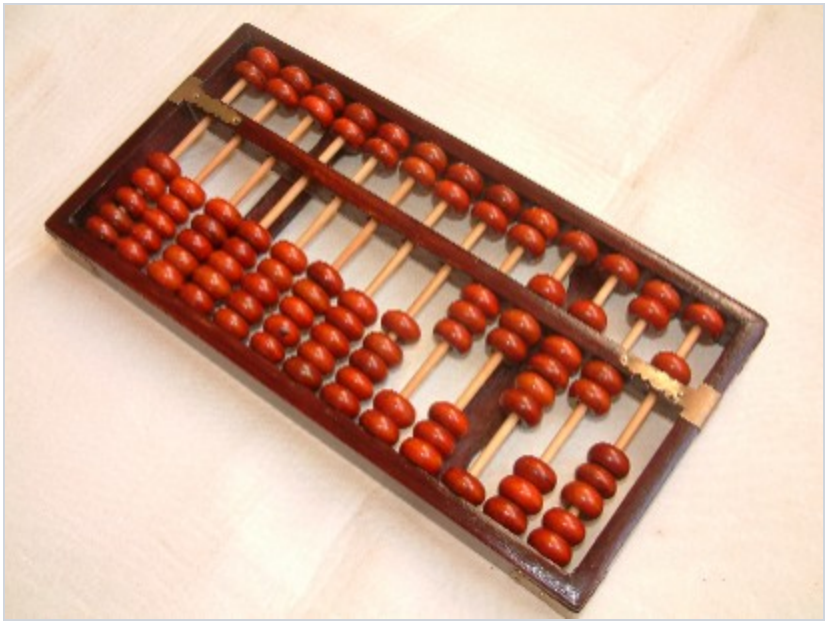


Figure 1.1. *A traditional Chinese suanpan abacus. The same instrument, in essence, was used for accounting in Mesopotamia, Egypt, Greece, Rome, China, Japan, and across the Islamic world.* en.wikipedia.org/wiki/Abacus

The abacus is the oldest surviving computing device. By moving beads along rods, a trained operator could perform addition, subtraction, multiplication, division, and even, with practice, square roots, at speeds that competed with early electronic calculators well into the 1940s. The abacus encodes numbers positionally, which is the same principle on which every modern digital computer rests. It is also, in a small but real sense, a stored-program device: the position of the beads represents both the operand and the intermediate state of the calculation.

The Antikythera Mechanism, circa 100 BCE



Figure 1.2. *The largest surviving fragment of the Antikythera mechanism, recovered in 1901 from a Roman-era shipwreck off the Greek island of Antikythera. Built around 100 BCE, it modelled the motions of the sun, moon, and planets with thirty-odd bronze gears.* en.wikipedia.org/wiki/Antikythera_mechanism

For nineteen centuries the Antikythera mechanism sat in a chest of bronze fragments at the bottom of the Aegean. When it was finally CT-scanned in the 2000s, the world discovered that ancient Greece had possessed an analog computer: a hand-cranked geared device that predicted eclipses, tracked the Olympic Games on a four-year cycle, and modelled the apparent motion of five planets. There is nothing else like it from antiquity. Whatever industrial culture produced it left no other example, and the line of mechanical computing fell silent for a thousand years.

Pascal, Leibniz, and the Mechanical Calculators

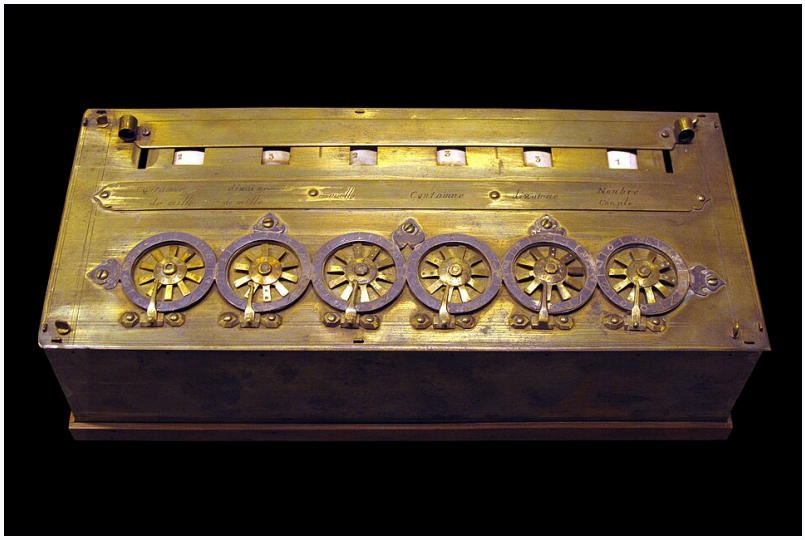


Figure 1.3. *The Pascaline, built by Blaise Pascal in 1642.* Pascal's calculator

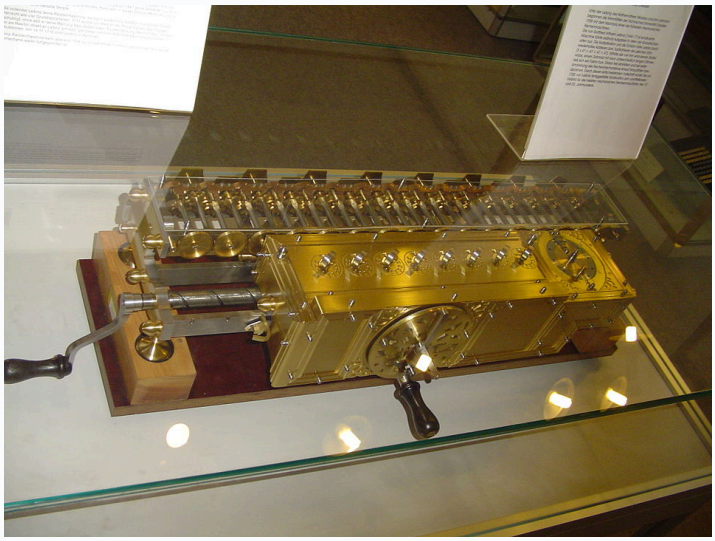


Figure 1.4. *Leibniz's Stepped Reckoner, 1673.* Stepped reckoner

In 1642, the nineteen-year-old Blaise Pascal built a mechanical calculator to help his father, a tax commissioner in Rouen, perform the arithmetic of public accounting. Three decades later, Gottfried Wilhelm Leibniz extended the idea with the *stepped reckoner*, which could multiply and divide. Leibniz also wrote, in 1703, the paper *Explication de l'Arithmétique Binaire*, in which he set out the binary number system, three hundred years before any device would exploit it.

Babbage, Lovelace, and the First Programs

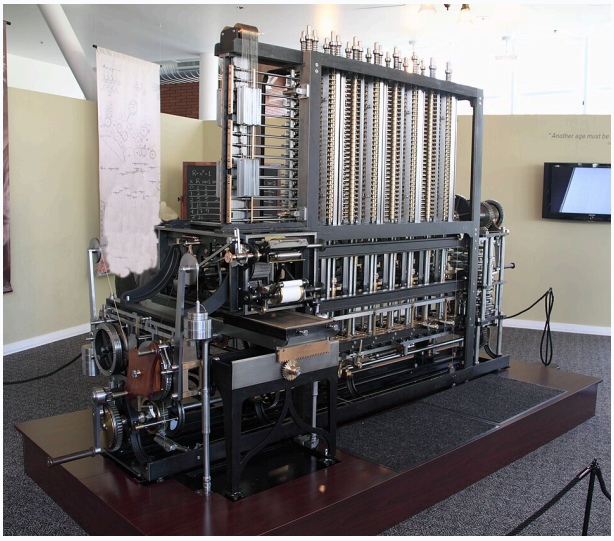


Figure 1.5. *Charles Babbage's Difference Engine No. 2, completed (posthumously) at the London Science Museum in 1991.* Difference engine



Figure 1.6. *Ada Lovelace, painted by Margaret Sarah Carpenter, 1836.* Ada Lovelace

In the 1830s, the English mathematician Charles Babbage designed a machine he called the *Analytical Engine*, a programmable, general-purpose mechanical computer driven by punched cards. He never finished it; the engineering tolerances of Victorian England were not equal to the task, and his money ran out. But the design was complete, and one of his collaborators, Ada Lovelace, the daughter of the poet Lord Byron, wrote a sequence of operations for the Engine to compute Bernoulli numbers. It was, by any reasonable definition, the first computer program, written for a machine that did not yet exist and would not for another century.

Lovelace also saw, more clearly than Babbage himself, what the device implied. Machines, she wrote, could in principle manipulate symbols, not just numbers, and might

one day compose music or assist in any field "in which the relations of things susceptible of being expressed by abstract signs are involved." Every program you will ever write rests on the truth of that paragraph.

→ **FURTHER READING: BEFORE ELECTRONICS**

- History of computing hardware *Wikipedia, comprehensive overview*
- The Computer History Museum timeline *computerhistory.org*
- Ada Lovelace's notes on the Analytical Engine *Finding Ada*
- Ada Lovelace: The First Computer Programmer *BBC podcast*

1.1 The Universal Machine

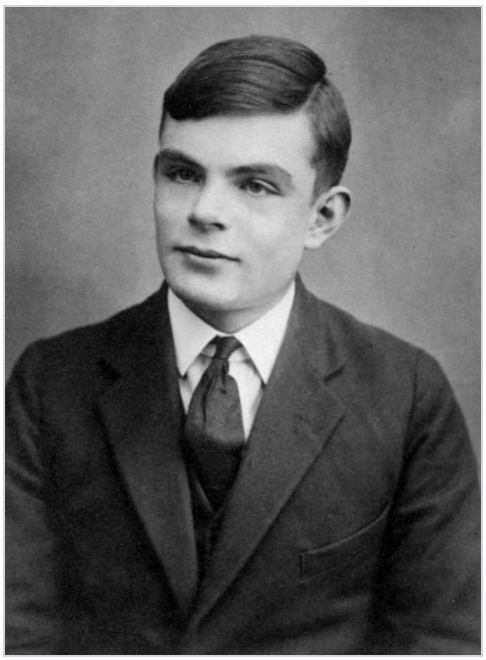


Figure 1.7. *Alan Turing, aged 16, in 1928.*

In 1936, a twenty-four-year-old English mathematician named Alan Turing published a paper with the unpromising title *On Computable Numbers, with an Application to the Entscheidungsproblem*. It is among the most consequential documents ever written. In it, Turing described a device that did not exist (a mathematical fiction, a thought experiment) that could, given the right instructions, simulate any other computing device imaginable. He called it an "a-machine"; we now call it a **Turing machine**.

The device was preposterously simple. It consisted of an infinite tape divided into cells, a head that could read or write one symbol at a time, and a small table of rules of the form: *if you see this symbol in this state, write that symbol, move left or right, and change state*. From this minimum, Turing proved, you could compute anything that was computable at all.

The point is not the tape. The point is what Turing's argument established: that **there exists a single kind of machine which, by being fed different instructions, can perform the work of any other kind of machine**. This is the idea of the *universal computer*, and every device you own (phone, laptop, server, watch) is one of them. They differ only in speed, capacity, and what is plugged in.

HISTORY The word *computer* originally referred to a person, usually a woman, whose job was to perform calculations by hand. NASA's "human computers", including Katherine Johnson, Dorothy Vaughan, and Mary Jackson, continued to work into the 1970s. The machine took the name from the role it replaced.

→ **FURTHER READING: TURING & COMPUTABILITY**

- Turing's original 1936 paper (PDF) *University of Virginia*
- Turing Machines *Stanford Encyclopedia of Philosophy*
- Computerphile: Turing Machines Explained *YouTube*
- Andrew Hodges, *Alan Turing: The Enigma* *the definitive biography*

1.2 The von Neumann Architecture

In June 1945, the Hungarian-American polymath John von Neumann circulated a draft document, the *First Draft of a Report on the EDVAC*, describing the architecture of a computer that would store its program in the same memory as its data. This was a departure from the practice of the time, in which programs were entered by physically rewiring the machine. Once a program lived in memory, it could be modified, copied, transmitted, and executed without altering a single wire. The idea spread quickly, and within five years almost every computer being built used the design. We still call it the **von Neumann architecture**.



Figure 1.8a. *John von Neumann (1903–1957).*

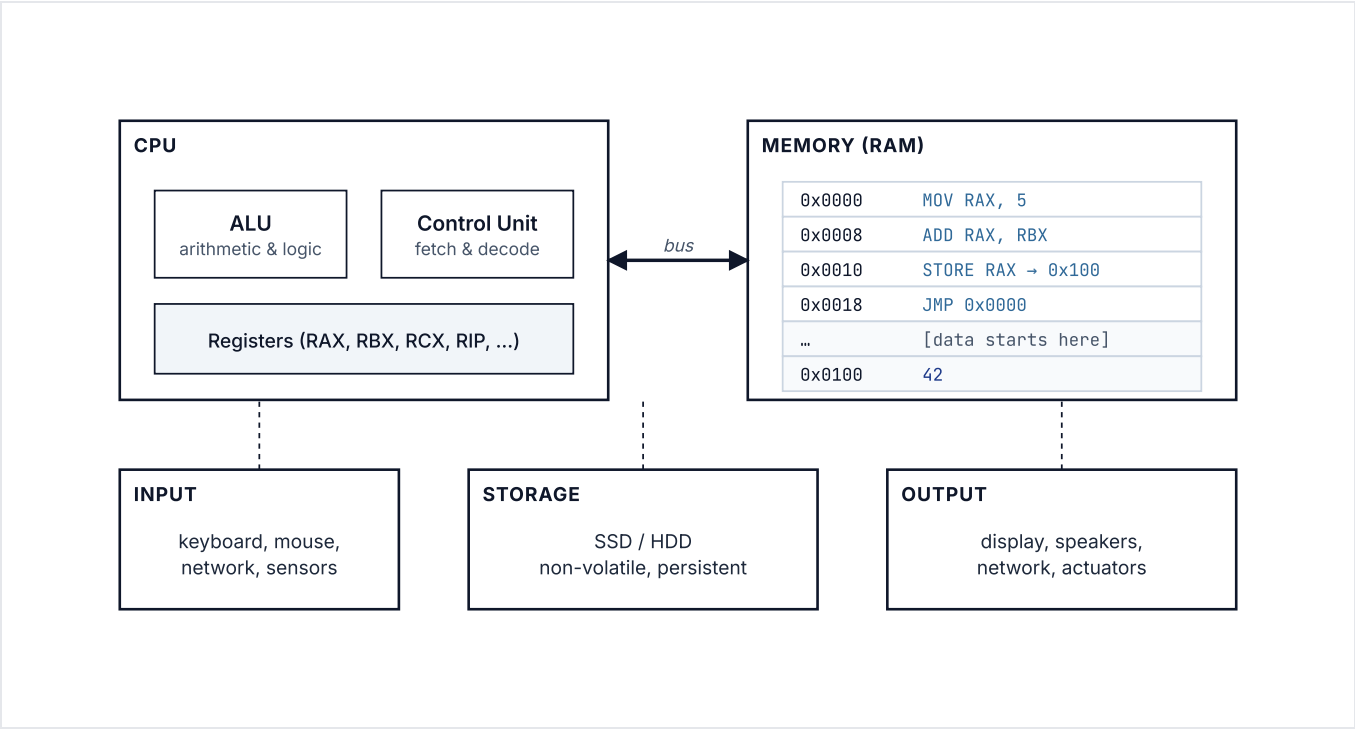


Figure 1.8. The von Neumann architecture. The CPU contains the arithmetic and control units and a small number of registers. Memory holds both instructions and data in the same address space. A bus carries values between the two. Input, storage, and output devices attach through controllers.

The architecture has four functional parts. The **CPU** is where computation happens. It contains an *arithmetic logic unit* for arithmetic and bitwise operations, a *control unit* for fetching and decoding instructions, and a small set of very fast working slots called *registers*. The **memory** (RAM, for random-access memory) holds programs and data; it is much larger than the registers but much slower. **Storage** (solid-state and rotating disks) holds data permanently across power cycles. **I/O devices** are the senses and limbs by which the machine reaches the world.

The CPU runs in a continuous loop, called the *fetch-decode-execute cycle*. It reads the instruction at the address held in a special register called the *instruction pointer*, decodes what it asks for, performs it, advances the pointer, and repeats. A modern processor performs this loop several billion times per second. The instruction at one address might be "add register A to register B"; the next might be "store the result at memory location 0x100"; the next might be "if register A is zero, jump to address 0x500". Out of these primitive moves, executed in sequence, every program runs.

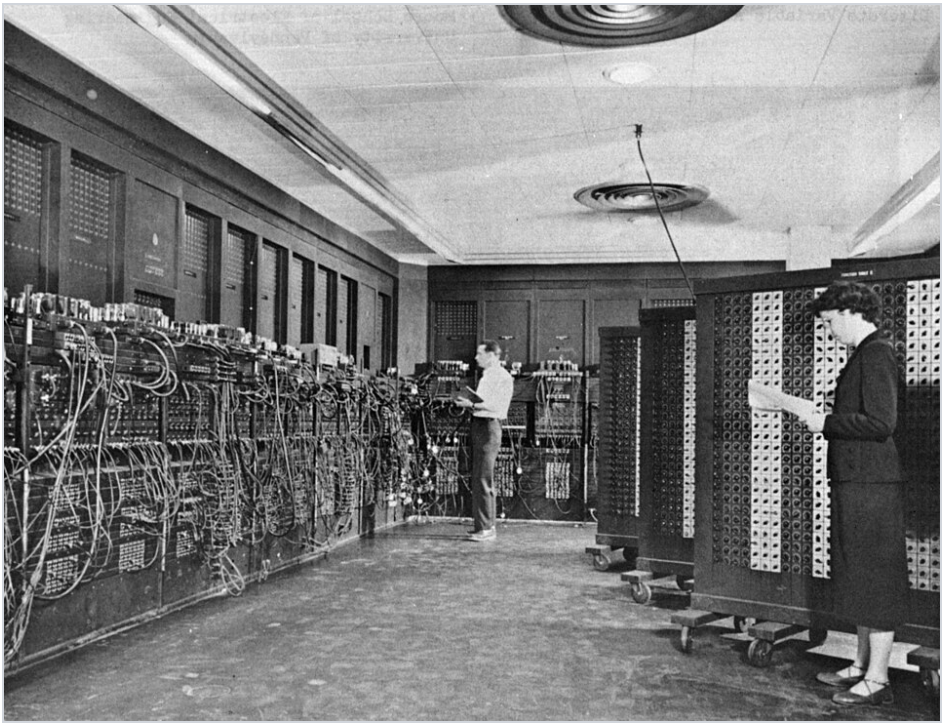


Figure 1.9. *ENIAC at the U.S. Army's Ballistic Research Laboratory, the first general-purpose electronic computer. 30 tonnes, 17,468 vacuum tubes, programmed by physically setting switches and re-routing cables. Shown are technicians Glen Beck and Betty Snyder; Snyder (later Betty Holberton) was one of ENIAC's original six women programmers.* ENIAC

DEF

Clock Cycle

The basic unit of time inside a CPU. A 3 GHz processor has three billion cycles per second; one cycle takes about 0.33 nanoseconds. A simple instruction may complete in one cycle; a complex one (a multiplication, a memory load) may take several. The number of cycles per instruction is called the CPI, and modern CPUs use pipelining and out-of-order execution to bring it below 1 in favourable code.

EXAMPLE 1.1 A single instruction, in three notations

Consider the operation: *add the integer in register RBX to the integer in register RAX*. Here it is, in three forms, descending toward what the silicon actually sees:

ASM

```
; assembly (what humans wrote in 1955)
ADD RAX, RBX
```

MACHINE

```
# machine code (what the CPU reads)
48 01 D8
```

BINARY

```
# the bits, in binary (what is on the wire)
01001000 00000001 11011000
```

The three lines describe the same operation. Every program in your computer's memory is, ultimately, a sequence of bytes like the second line. Assembly is a one-to-one human-readable transcription of it. Python is many, many layers above.

→ **FURTHER READING: COMPUTER ARCHITECTURE**

- *Computer Systems: A Programmer's Perspective* (Bryant & O'Hallaron) *the standard university text*
- Nand to Tetris *build a computer from logic gates, free course*
- Crash Course Computer Science *YouTube, 40 episodes*
- Von Neumann architecture *Wikipedia*
- Charles Petzold, *Code: The Hidden Language of Computer Hardware and Software* *superb introduction from first principles*

1.3 The Memory Hierarchy

Memory in a modern computer is not one thing. It is a hierarchy of several layers, each faster and smaller than the next. Understanding this hierarchy, and the orders of magnitude that separate its levels, is the difference between writing programs that are fast and programs that pretend to be.

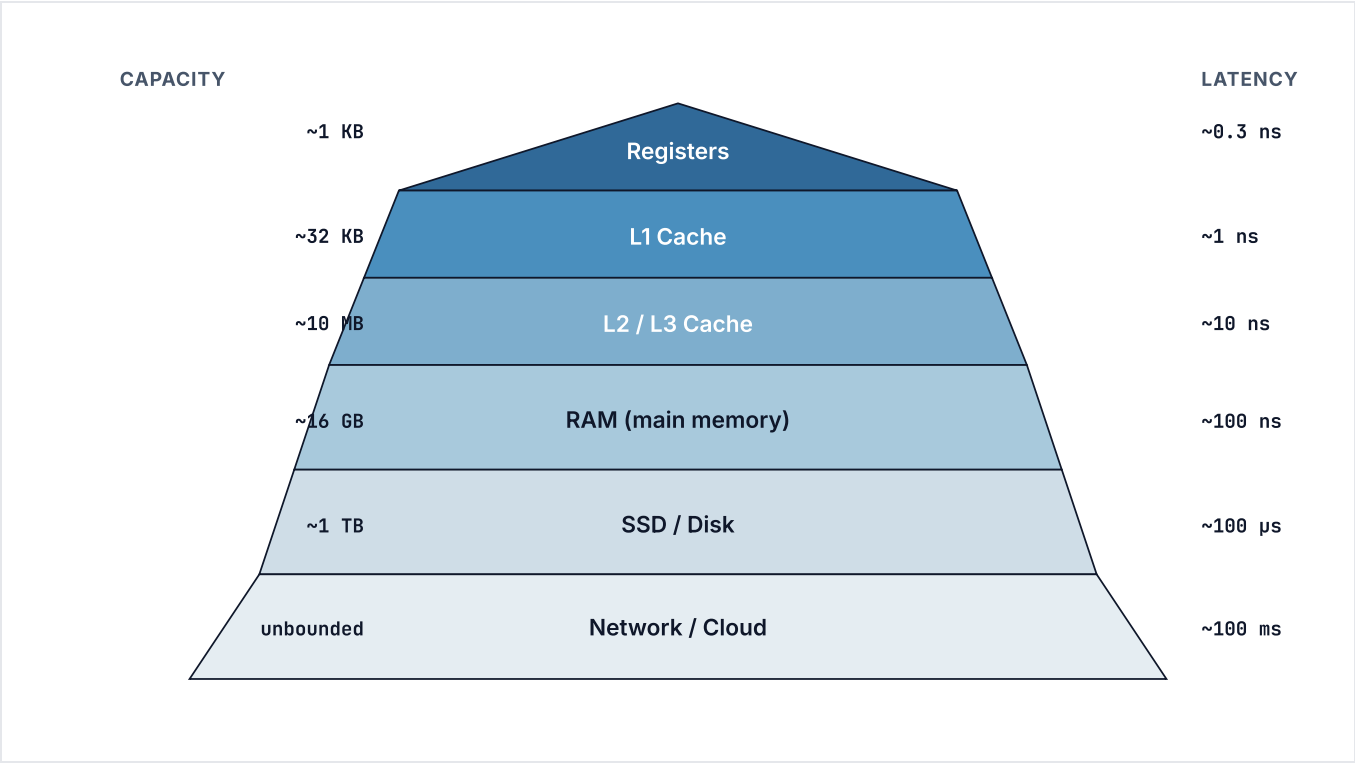


Figure 1.10. The memory hierarchy of a contemporary laptop. Each level is roughly an order of magnitude slower than the one above it. The numbers are approximate, but the ratios are not.

To grasp these numbers, multiply each by ten billion to scale the timings into human ones. A register access becomes about 3 seconds. An L1 cache access becomes 10 seconds. An L2 / L3 access becomes a minute and a half. A RAM access becomes about 15 minutes. An SSD read becomes a little over a day. A network round-trip becomes about a month. *This is the world your program lives in.* When people say a Python program is "slow", they almost always mean it is spending too much time at the lower levels of this pyramid, waiting for memory, for disk, for the network, and not enough time near the top.

PERFORMANCE INTUITION If a program runs a loop of arithmetic over values already in cache, it can manage billions of operations per second. If the same loop has to fetch each value from RAM, the rate drops by a factor of 100. If it has to read each value from SSD, the rate drops by another factor of 1,000. The size of these gaps is why so much modern software design is, in the end, about *locality*: keeping the data near the work.

→ FURTHER READING: MEMORY & CACHING

- Ulrich Drepper, "What Every Programmer Should Know About Memory" (PDF) *the definitive technical guide, 2007*
- Latency Numbers Every Programmer Should Know *Jeff Dean's famous list*
- Latency Numbers: Interactive Visualisation *how the numbers have changed over decades*

1.4 Numbers in Binary

The CPU only knows numbers, and even those it represents in binary, a positional notation with base 2 instead of base 10. Each digit in a binary number is called a *bit* and takes the value 0 or 1. Eight bits form a *byte*, which can represent $2^8 = 256$ distinct values.

The shift from decimal to binary is mostly a mechanical translation. The decimal number 13, written 1101 in binary, decomposes as:

WORKING

$$
\begin{aligned}
&1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
=\ &8 + 4 + 0 + 1 \\
=\ &13
\end{aligned}
$$

Integers: Two's Complement

Storing negative numbers is the first complication. Almost every modern CPU uses a scheme called *two's complement*. In an 8-bit two's-complement representation, the leftmost bit signals sign (0 for positive, 1 for negative), but the negative values are not simply the positives with the sign flipped. Instead, to negate a number, you invert all the bits and add 1.

Decimal	8-bit Binary
0	`00000000`
1	`00000001`
127	`01111111`
-1	`11111111`

-2	`11111110`
-128	`10000000`

Why this strange-looking scheme? Because under two's complement, the CPU can use a single circuit, the addition circuit, to perform both addition and subtraction. Computing `5 - 3` is mechanically identical to computing `5 + (-3)`. The cost of this elegance is a small asymmetry: an 8-bit two's-complement integer ranges from -128 to +127, not -127 to +127. One negative number has no positive partner.

EXAMPLE 1.2 Integer overflow, in C and in Python

Most languages treat an integer as a fixed-size container. C's `int` is typically 32 bits, and what happens when you exceed its range is among the most quietly devastating bugs in programming history:

C

```c
// C
int a = 2147483647;     // 2^31 - 1, the maximum 32-bit signed int
int b = a + 1;
printf("%d", b);           // prints -2147483648  (wraps around!)
```

The 1996 explosion of the Ariane 5 rocket, which destroyed €500 million of European satellites, was caused by exactly this kind of overflow: a 64-bit floating-point value was converted into a 16-bit signed integer that could not hold it. Python's design rejects this trap entirely. Its integers grow as large as needed:

PYTHON

```python
# Python
a = 2 ** 1000
print(len(str(a)))          # 302: the number has 302 digits
print(a + 1 != a)           # True: no overflow, ever
```

OUTPUT

```
302
True
```

This is a trade. C's fixed-size integers are dramatically faster to manipulate; Python's are dramatically safer. You will, in your career, have reason to prefer each.

Real Numbers: IEEE 754

Storing non-integer numbers (3.14, 0.001, 6.022×10^{23}) requires a scheme more elaborate than counting in binary. The standard used by essentially every CPU built since the late 1980s is **IEEE 754**. It represents a real number as a triple: a sign bit, an exponent, and a fraction (also called the mantissa or significand).

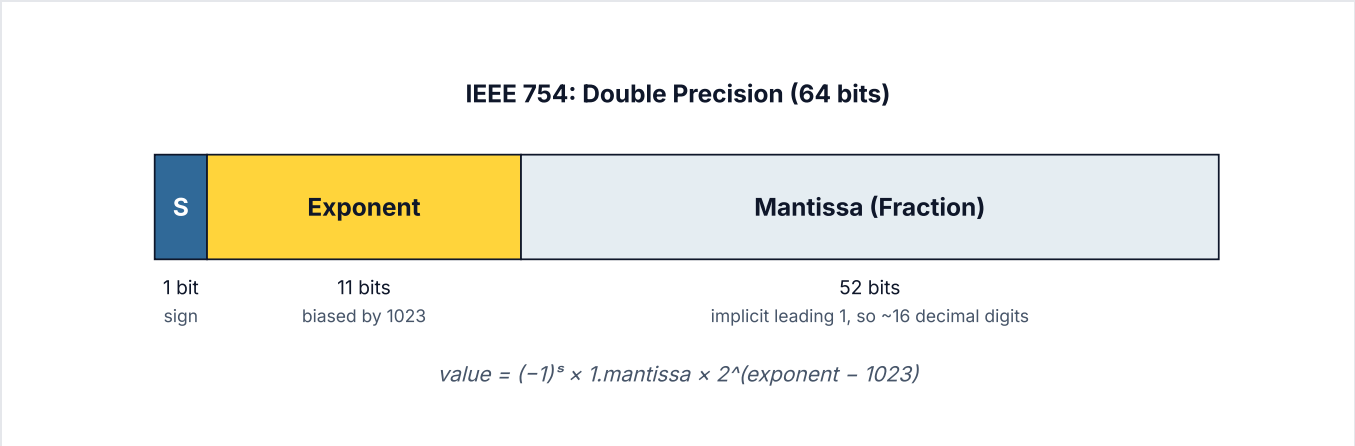


Figure 1.11. The 64-bit float layout. A 32-bit float is the same scheme with smaller fields (1 + 8 + 23). Python's `float` is always 64-bit.

The consequence of this representation is a fact every programmer must internalise: **floating-point numbers are approximations**. The decimal 0.1 cannot be expressed exactly in binary, much as 1/3 cannot be expressed exactly in decimal. The closest 64-bit IEEE 754 approximation of 0.1 is, in fact:

EXACT VALUE

```
0.1000000000000000055511151231257827021181583404541015625
```

This tiny error, harmless in isolation, accumulates over millions of operations and reveals itself in unexpected places. We will return to the consequences in Chapter 4. For now, internalise the diagram and the principle.

→ **FURTHER READING: BINARY & IEEE 754**

- The Floating-Point Guide *floating-point-gui.de, an excellent primer*
- Python docs: Floating Point Arithmetic, Issues and Limitations *official, concise*
- David Goldberg, "What Every Computer Scientist Should Know About Floating-Point Arithmetic" *the canonical deep dive*
- Two's complement *Wikipedia, with worked examples*

1.5 Machine Code, Assembly, and the First Bug

A program in memory is a sequence of bytes interpreted as instructions. Each instruction tells the CPU to do one of a few hundred primitive operations: load a value, store a value, add, subtract, compare, jump. The encoding is determined by the CPU's *instruction set architecture*: x86-64 for most laptops and desktops, ARM64 for phones and recent Macs, RISC-V for an increasing slice of embedded and research hardware.

Direct programming in machine code was abandoned by the late 1950s. Programmers wrote in *assembly*, a near-perfect one-to-one human-readable transcription of machine code, until the high-level languages of the 1960s displaced it for most work. Today, assembly remains in use only for the innermost loops of operating-system kernels, drivers, and certain performance-critical libraries.

EXAMPLE 1.3 A C function and its assembly

This is the C function `add`:

```c
int add(int a, int b) {
    return a + b;
}
```

The x86-64 assembly it compiles to under optimisation looks like this:

```asm
add:
    lea     eax, [rdi + rsi]   ; eax = rdi + rsi
    ret                        ; return; result is in eax
```

Two instructions. The C function's arguments arrive in registers `rdi` and `rsi` (per the System V calling convention), the sum is placed in `eax`, and `ret` hands control back to the caller. The equivalent Python function, also a one-liner, runs through perhaps two hundred machine instructions on the interpreter side before the addition itself is performed.

The First Bug

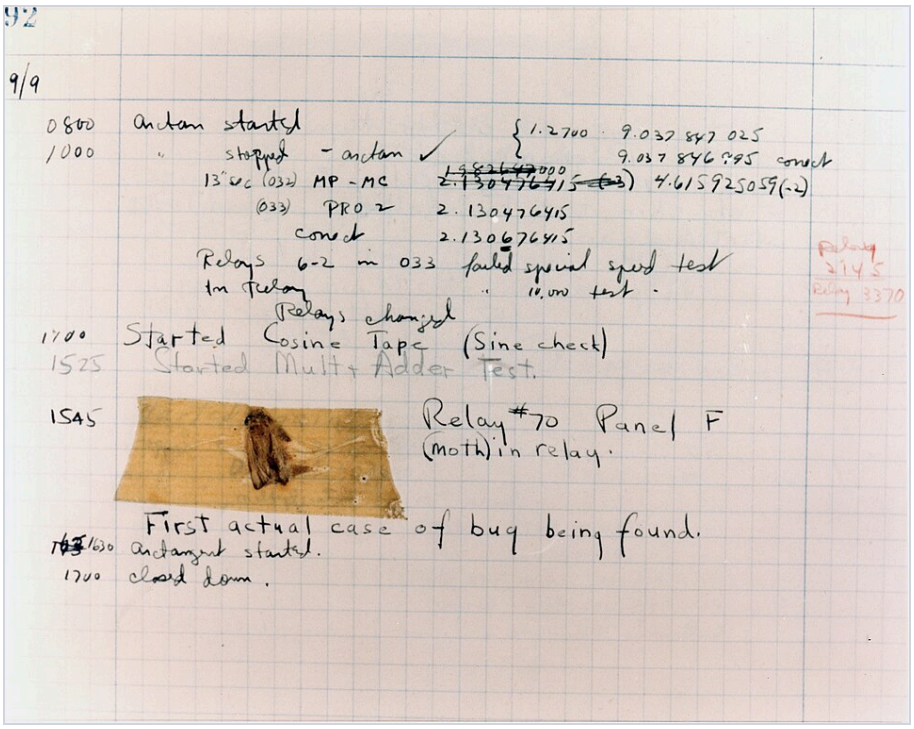


Figure 1.12. *The log entry from the Harvard Mark II computer team, 9 September 1947: "First actual case of bug being found." A moth had become trapped in a relay; Admiral Grace Hopper's team taped it into the logbook. The word "bug" had been used informally by engineers for decades, but this entry made it permanent.* Etymology of "bug"

The Call Stack

When a function calls another function, the CPU must remember where it was so it can come back. The mechanism is the *call stack*: a region of memory that grows downward as functions are entered and shrinks upward as they return. Each function's local variables, parameters, and return address live in a region of the stack called its *stack frame*.

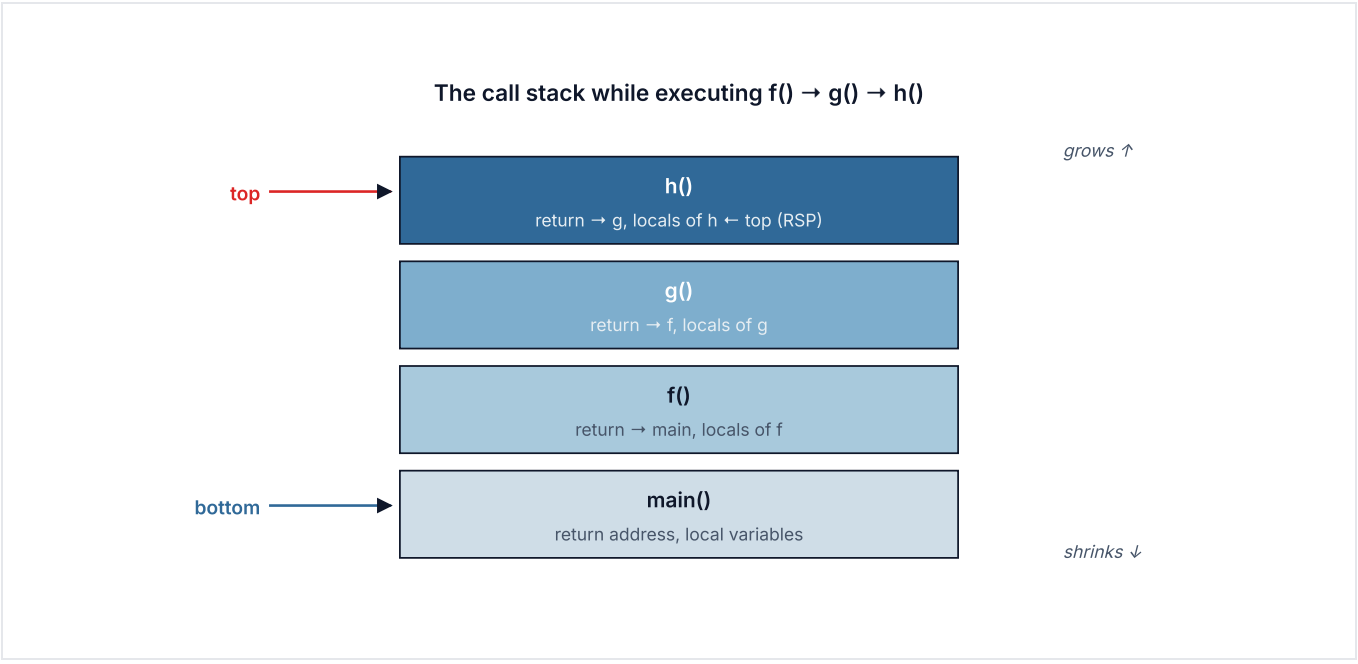


Figure 1.13. The call stack. Each function call pushes a new frame onto the top of the stack; each return pops one off. The stack is finite (typically a few megabytes), and a program that recurses without bound will eventually exhaust it (a stack overflow).

Python, too, has a call stack. When an exception is raised, the *traceback* it prints is, literally, a snapshot of this stack at the moment of the error. We will see this again in Chapter 21.

→ **FURTHER READING: ASSEMBLY & THE STACK**

- Compiler Explorer (godbolt.org) *type C, see the assembly; invaluable*
- Harvard CS61: Systems Programming and Machine Organization *free online course materials*
- Jonathan Bartlett, *Programming from the Ground Up* *assembly for beginners*

1.6 Processes, Threads, and the Operating System

Real computers do not run one program at a time. They run dozens, sometimes hundreds: your editor, your web browser, your music player, the daemons that keep your network alive, the system that updates your clock. The illusion that each program owns the machine is sustained by a piece of software called the *operating system*.

An operating system (Windows, macOS, Linux, iOS, Android) does three principal things. It *multiplexes* the CPU, slicing time into small intervals, running a different program in each, switching between them faster than any human perception can detect. It *isolates* programs from one another, giving each its own private view of memory, called a *virtual address space*, so that one program's bug cannot corrupt another's data. And it *mediates* access to hardware: your program does not write directly to the screen or read from the network card; it asks the operating system, which checks permissions and routes the request.

DEF **Process**

A running instance of a program, as seen by the operating system. A process has its own memory space, its own file descriptors, its own permissions, and at least one thread of execution. When you launch Python, the OS creates a Python process. When you launch a second Python interpreter, you get a second, completely independent process.

DEF **Thread**

A path of execution within a process. A single process may have many threads, sharing the same memory but each with its own call stack and its own current instruction pointer. Threads are how a program does several things in parallel. They are also a notorious source of bugs, because shared memory plus parallel execution is a difficult combination to reason about.

Python has a famously specific relationship with threads: the *Global Interpreter Lock* (GIL) ensures that only one thread inside a CPython interpreter executes Python bytecode at a time, even on a multi-core machine. This trade, simplicity of the interpreter's internals at the cost of CPU-parallelism, has been a subject of debate for thirty years. We will look at it properly in Chapter 21. For now, just know it is there.

→ **FURTHER READING: OPERATING SYSTEMS**

- *Operating Systems: Three Easy Pieces* *free book, the modern standard*
- MIT 6.828 lectures *YouTube, advanced*
- Operating system *Wikipedia overview*
- What Is the Python Global Interpreter Lock? *Real Python*

1.7 What Python Inherits From All This

The Python you write does not, in any literal sense, run on the machine described above. It runs on a virtual machine (a piece of C code called the CPython interpreter) which itself runs on the machine described above. But the constraints of the underlying hardware leak through. When you create a list of a million integers, those million PyObject pointers occupy real bytes in real memory. When you call a function, a real stack frame is pushed onto a real stack. When your program is slow, it is slow for reasons rooted in the hierarchy of Figure 1.10.

A programmer who knows none of this can still write Python. They will, however, be regularly puzzled. Why is appending to a list fast but inserting at the front slow? Why is dictionary lookup constant-time but list `in` linear? Why does `multiprocessing` exist when `threading` already does? Each answer involves the hardware. We will encounter all

three before this book is finished, and you will understand each immediately, because you read this chapter.

The remaining chapters of Part I cover the second half of the foundation: where Python itself came from, what CPython actually is, and how a Python statement becomes electricity. After that the language itself begins.

§ § §

Exercises

EXERCISE 1.1 Bit counting

How many distinct values can be represented by 1 bit? By 4 bits? By 16 bits? By 64 bits? Express each as a power of 2 and approximate it in decimal. Why does a 64-bit integer suffice for almost every practical counting problem?

EXERCISE 1.2 Two's complement by hand

Compute the 8-bit two's-complement representation of −37. Show your working: write +37 in binary, invert the bits, then add 1. Verify by adding your result to +37 in binary and observing that the answer is zero (with a discarded carry).

EXERCISE 1.3 Memory hierarchy scaling

If a register access takes 0.3 nanoseconds and a network round-trip takes 100 milliseconds, what is the ratio between them? If you scaled "register = 1 second", what would a network round-trip be, in human time?

EXERCISE 1.4 Why 0.1 is not 0.1

Explain in your own words, with reference to Figure 1.11, why a 64-bit float cannot represent the decimal number 0.1 exactly. What other simple decimal numbers *can* it represent exactly? (Hint: which fractions have terminating expansions in base 2?)

EXERCISE 1.5 Stack overflow

If a function call pushes a stack frame of 200 bytes onto a 1-megabyte call stack, roughly how deep can the recursion go before the stack overflows? Compare your answer to Python's default recursion limit (1000). Why might Python's limit be lower than the hardware allows?

EXERCISE 1.6 An instruction budget

A modern CPU executes around 3×10^9 instructions per second per core. A typical web page takes around 100 milliseconds to render. Estimate, very roughly, how many CPU instructions are spent rendering one web page. Does this match your intuition about how complex a web page is?

EXERCISE 1.7 Processes vs threads

Describe, in plain language, the difference between a process and a thread. Give one situation in which you would prefer threads (many tasks sharing data), and one in which you would prefer processes (many tasks that must not interfere with each other).

EXERCISE 1.8 A research question

The Ariane 5 rocket explosion of 4 June 1996 was caused by an integer overflow. Spend twenty minutes researching what happened. Write a one-paragraph technical summary in your own words. The point of this exercise is not the rocket; the point is that even very expensive teams sometimes write code without thinking about the hardware constraints we have discussed.

Solutions to Chapter 1 Exercises

Solution 1.1: Bit counting

1 bit gives $2^1 = 2$ values (0 or 1). 4 bits give $2^4 = 16$. 16 bits give $2^{16} = 65{,}536$. 64 bits give $2^{64} \approx 1.8 \times 10^{19}$, about 18 quintillion. A 64-bit integer can count every grain of sand on Earth and have room to spare; it can count the seconds since the universe began with about 22 orders of magnitude unused. The only domains that need more are mathematics (cryptography, combinatorics) and certain scientific simulations.

Solution 1.2: Two's complement of -37

WORKING

```
+37  =  00100101
invert: 11011010
add 1:  11011011

verify: 00100101    (+37)
      + 11011011    (-37)
      = 00000000    (carry discarded)
```

So −37 in 8-bit two's complement is `11011011` (decimal 219 if interpreted as unsigned).

Solution 1.3: Memory hierarchy scaling

The ratio is $0.1 / (0.3 \times 10^{-9}) \approx 3.3 \times 10^8$, about a third of a billion times slower. Scaled so a register access is 1 second, a network round-trip is roughly 333 million seconds, which is about 10.5 years. This is why caching matters so much: the cost of going to the network for data you already have nearby is, on this scale, geological.

Solution 1.4: Why 0.1 is not 0.1

A binary fraction terminates only if the denominator (when the fraction is in lowest terms) is a power of 2. The fraction 1/10 = 1/(2 × 5) has a factor of 5 in its denominator, so it does not terminate in binary; it produces a repeating expansion 0.000110011001100... that must be truncated to fit in 52 mantissa bits. Decimals that *do* terminate include 0.5 (= 1/2), 0.25 (= 1/4), 0.125 (= 1/8), and any sum of these. So 0.5 + 0.25 = 0.75 is exact in a float; 0.1 + 0.2 = 0.3 is not.

Solution 1.5: Stack overflow

1 MB / 200 B = 5,242 frames at most. Python's default limit of 1000 is set well below the hardware ceiling, deliberately, to give a clearer error message before the C stack itself overflows, which would crash the interpreter outright. The limit can be raised with

`sys.setrecursionlimit()`, but doing so without an architectural reason is usually a code smell. Most deep recursions are better rewritten as loops.

Solution 1.6: An instruction budget

$3 \times 10^9 \times 0.1 = 3 \times 10^8$ instructions per page render, three hundred million. This is for one core; modern machines have several. The number sounds enormous and is. The amount of work a modern browser does to lay out a page (parsing HTML, resolving CSS, computing layout, painting pixels, running JavaScript) comfortably fills it.

Solution 1.7: Processes vs threads

A *process* is a running program with its own private memory space; an operating system isolates each from the others. A *thread* is a path of execution within a single process; multiple threads in one process share memory directly. Threads are preferable when tasks need to communicate constantly (a web server handing requests off to worker threads that share a connection pool). Processes are preferable when tasks must not be able to corrupt each other (a browser's tab-per-process model, where a crash in one tab does not bring down the others).

Solution 1.8: Ariane 5

The Ariane 5 inherited inertial reference software from Ariane 4. In a routine pre-flight conversion, a 64-bit floating-point measurement of horizontal velocity was cast into a 16-bit signed integer. Ariane 5 was a faster rocket; the velocity exceeded 32,767 (the maximum value the 16-bit integer could hold); the conversion overflowed and produced a value the guidance system interpreted as garbage. The rocket veered, broke up under aerodynamic stress, and self-destructed 37 seconds after launch. The original Ariane 4 conversion had been deemed safe based on physical limits the older rocket could not exceed; nobody re-verified that assumption when the software was carried over. The fault was not a bug in the conversion code. The fault was the absence of a check on the assumption beneath it.

CHAPTER 2 · PART I · THE GROUND

The Lineage of Python

No language is invented from nothing. Python is the considered descendant of half a dozen earlier languages, the personal project of one programmer, and the collective work of thousands since. This chapter traces where it came from and explains what actually happens inside the program that runs your code.

> *I decided to try to design a language that I myself wanted to use. I had no illusions that this would change the world, but I knew I wanted to scratch my own itch.*
>
> — Guido van Rossum, creator of Python

In December 1989, a thirty-three-year-old Dutch programmer named Guido van Rossum found himself with a week of free time over the Christmas holiday and a problem he had been turning over for years. His office at the Centrum Wiskunde & Informatica in Amsterdam was closed. He had a home computer and an idea: a scripting language that would have the power of the systems languages he used at work, but the gentleness of a teaching language he had helped build in the 1980s. He started writing an interpreter. He named the project after a British comedy troupe whose scripts he had been reading, *Monty Python's Flying Circus*. The snake came later, and only as a logo.

The language that grew out of that holiday is now, by most measures, the most widely taught and most widely written programming language in the world. To understand why it has the shape it does, you have to understand the languages it descended from, the decisions its creator made deliberately, and the machinery that the standard implementation, CPython, uses to bring your code to life. That is the work of this chapter.

2.1 A Family of Languages

By the time Python was conceived, programming languages had been evolving for more than thirty years, and a recognisable family tree had formed. Each new language was, in large part, a reaction to the frustrations of the ones before it. Knowing the broad shape of that tree explains a surprising amount about why Python looks the way it does.

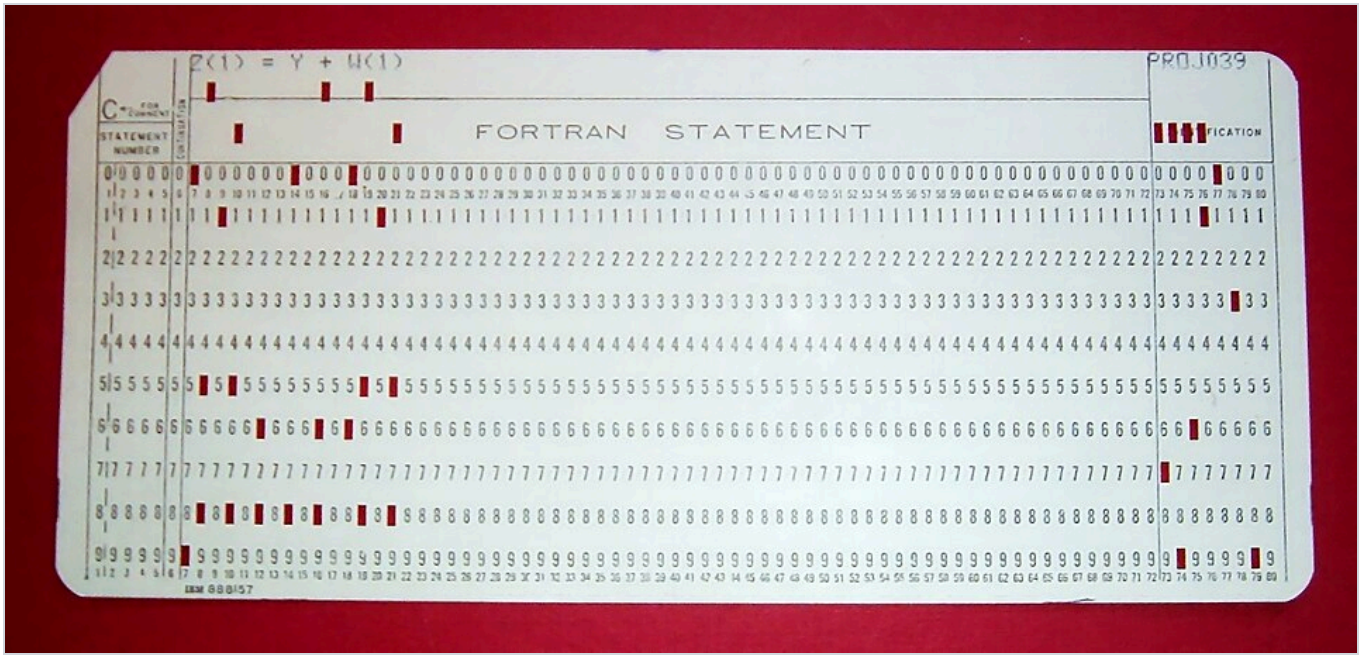


Figure 2.1. *A FORTRAN punched card. Into the 1970s, a program was a stack of these cards, one statement per card, fed into a reader. A dropped deck meant a re-sorted afternoon.* The punched-card era

The first high-level language to see wide use was **FORTRAN** (1957), built by a team under John Backus at IBM. Its great contribution was the *compiler*: a program whose job was to translate something resembling mathematical notation into machine code automatically, so that programmers no longer had to write the machine code themselves. Many doubted a compiler could ever produce code efficient enough to be worth using. Backus proved them wrong, and in doing so invented an entire category of software.

The languages that followed each added an idea. **LISP** (1958) introduced the notion that code and data could share the same form, and brought recursion and automatic memory management into the mainstream. **ALGOL** (1960) gave us block structure and the lexical scope rules that nearly every later language, including Python, would adopt. **Simula** (1967) invented the *object*: a bundle of data together with the procedures that act on it. **C** (1972) struck a balance between closeness to the hardware and human readability so successful that it became, and remains, the language in which operating systems are

written. **Smalltalk** (1972) took Simula's object idea to its logical conclusion, making everything in the language an object, an idea Python would later embrace wholeheartedly.

→ FURTHER READING: THE FAMILY OF LANGUAGES

- History of programming languages *Wikipedia overview*
- The History of Programming Languages *talk by various, computer history*
- HOPL: History of Programming Languages conference papers *the primary academic record*

2.2 The Three Parents of Python

Python did not descend evenly from the whole tree. Three languages shaped it directly, and each left a different mark.

ABC: the gift of readability

Through most of the 1980s, Guido van Rossum worked at CWI on a language called **ABC**, intended to teach programming to beginners and to be usable by scientists who were not professional programmers. ABC was where Python's most distinctive feature was born: the use of indentation, rather than braces or keywords, to mark which statements belong together. ABC also favoured a small number of powerful, high-level data types and an uncluttered syntax that read almost like English.

ABC ultimately failed to find an audience, and Guido came to understand why. It was a closed world. You could not easily add new features, talk to the operating system, or read and write ordinary files. It tried to be a complete environment rather than a tool that cooperated with the rest of the system. When Guido designed Python, he kept ABC's readability and threw out its isolation. Python would be extensible from the start, and it would talk to everything.

C: the language Python is written in

The standard Python interpreter is a program written in C, and this is no accident. C was the lingua franca of systems programming, the language in which the operating systems Python needed to run on were themselves built. Writing the interpreter in C meant Python could run anywhere C ran, which by 1990 meant almost everywhere. It also meant that

performance-critical pieces of Python, and later entire libraries such as NumPy, could be written in C and called from Python, giving the language a way to escape its own slowness when it mattered. We will return to this idea repeatedly; it is central to why a "slow" language powers so much fast software.



Figure 2.2. *Dennis Ritchie (1941–2011), creator of C.*

Modula-3: modules and exceptions

While at CWI, Guido encountered **Modula-3**, a language from the Digital Equipment Corporation research labs. From it Python borrowed its module system, the way code is organised into importable units, and the basic shape of its exception handling, the `try` and `except` machinery you will meet in Chapter 21. These were mature, well-considered designs, and rather than reinvent them Guido adapted what already worked.

> **DEF** **Scripting Language**
>
> A language designed for writing short programs that automate tasks, glue other programs together, or configure systems, rather than for building large standalone applications from scratch. The term has blurred over time: Python began as a scripting language and is now used to build systems of every size. The distinction was always more about intent than capability.

→ **FURTHER READING: PYTHON'S ORIGINS**

- Guido's Foreword for "Programming Python" (1996) *his own account of the early days*
- The History of Python *Guido's own blog on design decisions*
- ABC (programming language) *Wikipedia*

2.3 Guido's Design Decisions



Figure 2.3. *Guido van Rossum, who created Python over the winter of 1989 and led its development as "Benevolent Dictator For Life" until 2018.* Guido van Rossum

A language is the sum of thousands of small choices, and the choices that give Python its character were made early and held to with unusual consistency. Four of them matter most.

The first is **readability above cleverness**. Guido observed that code is read far more often than it is written: a line written once may be read a hundred times over the life of a program, by its author and by strangers. He optimised the language for the reader. This is why indentation is mandatory rather than decorative, why the language avoids cryptic punctuation, and why there is a cultural preference, captured in the Zen of Python, for there to be one obvious way to do a thing rather than many clever ones.

The second is that **everything is an object**. Numbers, strings, functions, classes, modules, even the types themselves: all are objects, with the same uniform interface. This decision, inherited from Smalltalk, gives the language a deep consistency. The whole of Chapter 3 is devoted to its consequences.

The third is **batteries included**. Python ships with a large standard library so that common tasks, reading files, parsing dates, fetching web pages, working with JSON, can be done out of the box without hunting for third-party code. This made Python immediately useful and helped it spread.

The fourth is **pragmatism over purity**. Where a purely object-oriented language insists you write `list.length()`, Python offers the blunter `len(list)` because it reads better. Where a purely functional language forbids changing data in place, Python allows it where it is convenient. Python borrows good ideas from every paradigm and refuses to be a slave to any one of them.

EXAMPLE 2.1 The Zen of Python

Hidden inside every Python installation is a short poem of design principles, written by Tim Peters in 1999. Type one line at the interpreter to read it:

PYTHON

```
import this
```

OUTPUT

```
The Zen of Python, by Tim Peters

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Readability counts.
...
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
```

The joke in the last line is about Guido, who is Dutch. The poem is not a specification, but when the community argues about whether a proposed feature belongs in the language, these nineteen aphorisms are quoted more often than any technical document.

→ **FURTHER READING: DESIGN PHILOSOPHY**

- PEP 20: The Zen of Python *the canonical text*
- PEP 8: Style Guide for Python Code *how Python is meant to be written*
- "Transforming Code into Beautiful, Idiomatic Python" by Raymond Hettinger *a classic talk on Pythonic style*

2.4 What CPython Actually Is

Here is a distinction that confuses many beginners and that, once clear, never confuses you again. **Python is a language specification.** It is a document describing what the language means: what `2 + 2` should produce, what a `for` loop does, how a function call behaves. **CPython is an implementation**: an actual program, written in C, that reads Python source code and executes it according to that specification. When you download "Python" from python.org, what you are installing is CPython.

CPython is by far the dominant implementation, so much so that its behaviour has, in places, become the specification by default. But it is not the only one, and the alternatives illuminate what an implementation is free to change.

Implementation	Written in	Distinctive trait
CPython	C	the reference; what almost everyone runs
PyPy	RPython	a JIT compiler, often several times faster
Jython	Java	runs on the JVM, interoperates with Java
IronPython	C#	runs on .NET
MicroPython	C	fits on a microcontroller with kilobytes of RAM
GraalPy	Java	runs on the GraalVM, high performance

The existence of PyPy is the most instructive. It runs the same Python language but, by compiling hot code to machine code as it runs (a technique called just-in-time compilation), it executes many programs several times faster than CPython. That CPython does not do this is a deliberate trade: a simpler interpreter that is easier to maintain, embed, and extend with C libraries, at the cost of raw speed. Throughout this book, unless stated otherwise, "Python" means CPython, because that is what you are almost certainly running.

→ **FURTHER READING: IMPLEMENTATIONS**

- The Python Language Reference *the specification itself*
- PyPy *the JIT-compiled implementation*
- CPython source code on GitHub *readable, well-organised, surprisingly approachable*

2.5 The Compilation Pipeline

Python is often called an interpreted language, and it is, but the description hides something. Before any interpretation happens, your source code passes through several stages of translation. Understanding these stages demystifies a great deal of the interpreter's behaviour, including why syntax errors are reported before your program runs at all, and what those `__pycache__` folders contain.

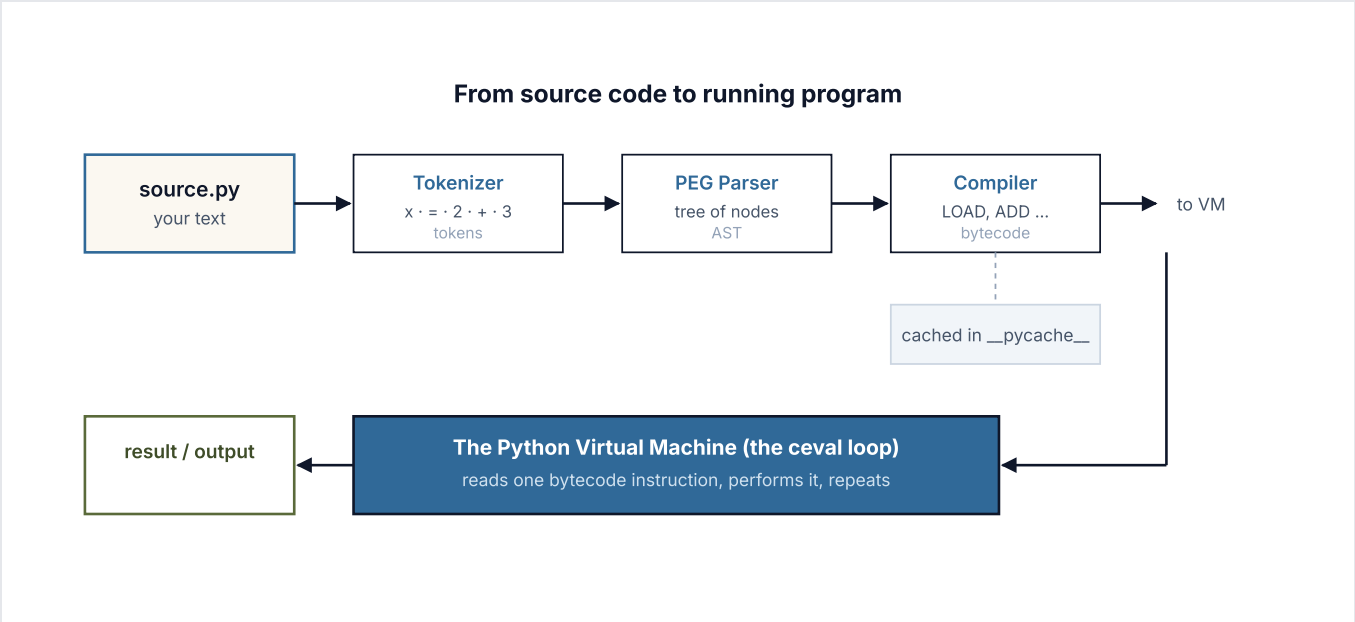


Figure 2.4. The CPython pipeline. The first three stages happen once, before execution; their result, the bytecode, is cached on disk so that unchanged modules need not be recompiled. The virtual machine then executes the bytecode instruction by instruction.

The **tokenizer** reads your source as a stream of characters and groups them into tokens: the name `x`, the operator `=`, the number `2`, and so on. The **parser** arranges those tokens into a tree that reflects the grammatical structure of the program, called the *Abstract Syntax Tree*. Since version 3.9, CPython uses a PEG parser, a more powerful design than the one it replaced. The **compiler** walks that tree and emits *bytecode*, a sequence of sim-

ple instructions for a virtual machine. Finally, the **virtual machine**, a loop in a C file historically named `ceval.c`, executes the bytecode one instruction at a time.

EXAMPLE 2.2 Looking at the bytecode

The `dis` module ("disassemble") lets you see the bytecode the compiler produced for any function. This is not a trick; it is the actual sequence the virtual machine will run.

PYTHON

```
import dis

def add(a, b):
    return a + b

dis.dis(add)
```

OUTPUT

```
  2           RESUME                   0
  3           LOAD_FAST                a
              LOAD_FAST                b
              BINARY_OP                0 (+)
              RETURN_VALUE
```

Five instructions. `LOAD_FAST` pushes a local variable onto an internal stack, `BINARY_OP` pops two values and pushes their sum, and `RETURN_VALUE` hands the result back. The Python virtual machine is a *stack machine*: nearly every instruction works by pushing values onto, or popping them off, an operand stack.

DEF **Bytecode**

An intermediate, low-level instruction set, simpler than Python but higher-level than machine code, that the Python virtual machine executes. It is portable across operating systems and CPUs, which machine code is not. The `.pyc` files in `__pycache__` are compiled bytecode, cached so that importing an unchanged module skips recompilation. They are an optimisation, safe to delete; Python regenerates them as needed.

→ **FURTHER READING: THE PIPELINE**

- CPython Developer's Guide: The Compiler *how it really works, from the source*
- The dis module documentation *every bytecode instruction explained*
- PEP 617: New PEG parser for CPython *why the parser was replaced in 3.9*

2.6 Reference Counting and the Garbage Collector

In a language like C, the programmer must explicitly free every piece of memory they allocate. Forgetting to do so is a *memory leak*; freeing memory still in use is a *use-after-free*, the source of a great share of the security vulnerabilities of the last forty years. Python relieves you of this burden entirely. You never free memory by hand. Understanding how it does this for you is worth a few pages, because it explains some otherwise mysterious behaviour.

CPython's primary mechanism is **reference counting**. Every object carries a hidden counter recording how many names, list slots, or other references currently point at it. Each new reference increments the count; each reference that goes away decrements it. The instant the count reaches zero, the object is unreachable, and its memory is reclaimed immediately.

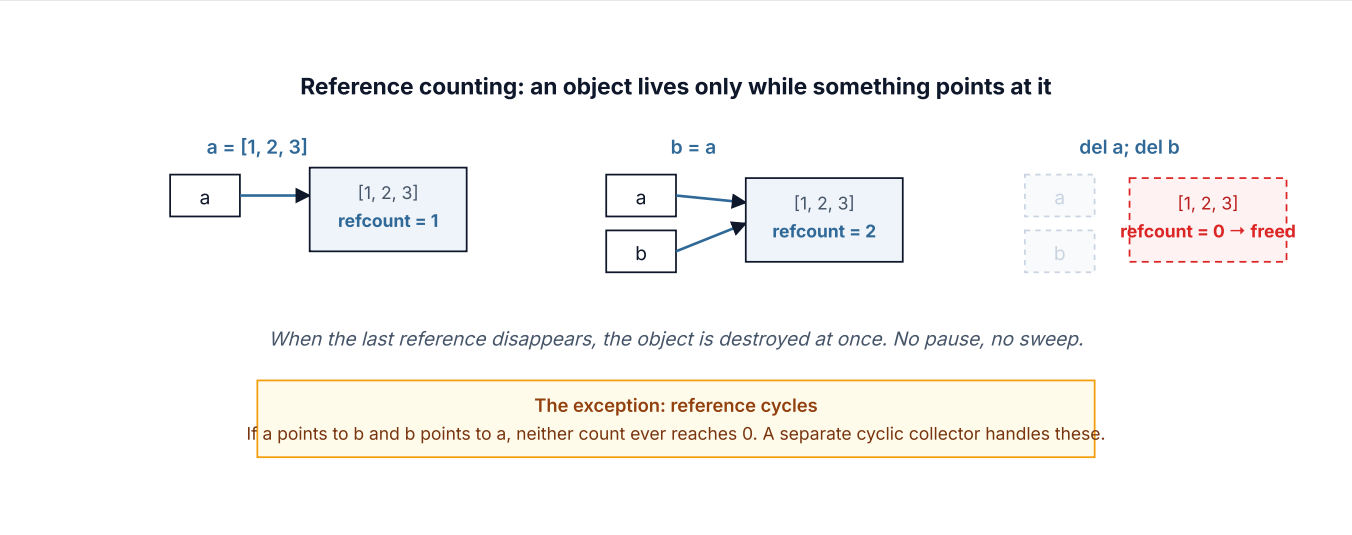


Figure 2.5. Reference counting in action. This is also the mechanism behind the aliasing behaviour of Chapter 3: when `b = a` makes two names point at one list, it is literally incrementing one shared counter.

Reference counting is simple and prompt, but it has one blind spot: *cycles*. If object A holds a reference to object B, and B holds a reference back to A, then even after every

outside name to them is gone, each still has a reference count of one, pointing at the other. Neither will ever be freed by counting alone. To catch these, CPython runs a second mechanism, a **generational cyclic garbage collector**, which periodically looks for groups of objects that refer only to each other and are reachable from nothing else, and reclaims them.

WHY TWO MECHANISMS Reference counting handles the overwhelming majority of objects instantly and predictably. The cyclic collector exists only to mop up the rare cycles that counting cannot. Splitting the work this way keeps the common case fast. You can inspect and even tune the cyclic collector through the `gc` module, though you will rarely need to.

→ **FURTHER READING: MEMORY MANAGEMENT**

- Python C-API: Reference Counting *Py_INCREF and Py_DECREF*
- CPython Developer's Guide: Garbage Collector *the cyclic collector in detail*
- Memory Management in Python *a clear, illustrated walkthrough*

2.7 The Global Interpreter Lock

No discussion of CPython is complete without the Global Interpreter Lock, universally abbreviated the GIL. It is the single most debated design decision in the language's history, and the one beginners most often hear about without understanding.

The GIL is a mutex, a lock, that ensures only one thread executes Python bytecode at any given moment, even on a processor with many cores. The reason it exists is reference counting. Recall that every object's reference count is constantly being incremented and decremented. If two threads tried to adjust the same count at the same time, the count could be corrupted, leading to objects freed too early or never freed at all. The GIL makes the interpreter's internals safe by the simplest possible means: it lets only one thread run Python code at a time, so the counts can never be touched concurrently.

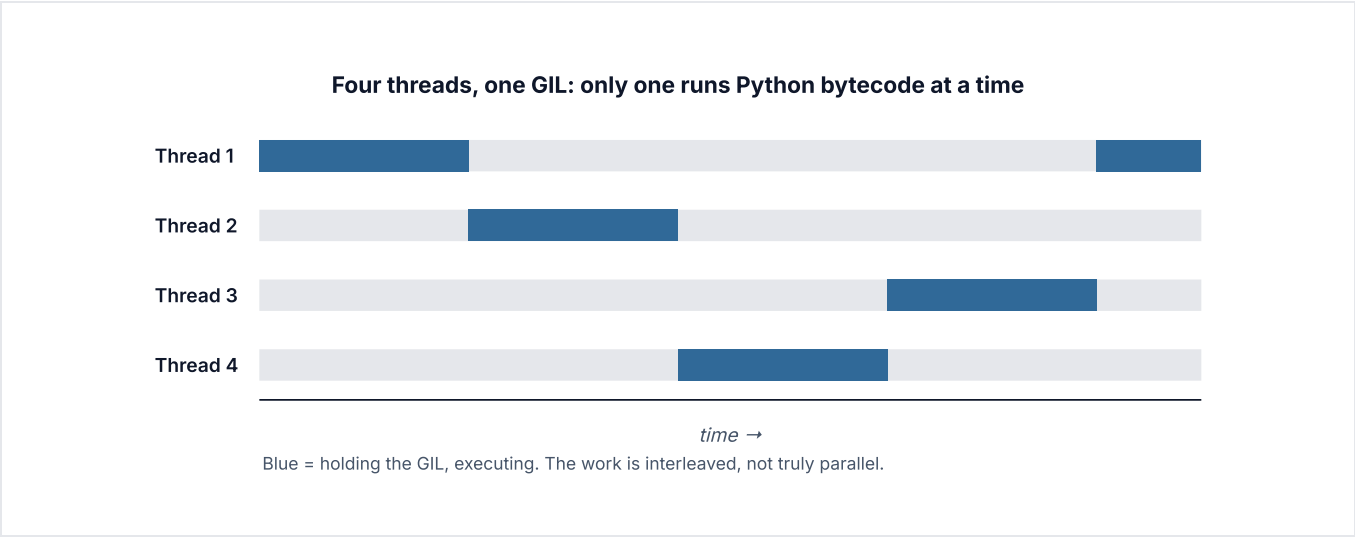


Figure 2.6. With the GIL, threads take turns. For work that spends its time waiting (network, disk), this is fine, because a waiting thread releases the GIL. For work that is pure computation, the threads cannot use multiple cores, and you reach instead for processes.

The consequence is precise and worth stating carefully. For work that is *input-output bound*, programs that spend most of their time waiting for the network, the disk, or a database, threads work well, because a thread that is waiting releases the GIL for others to use. But for work that is *CPU bound*, pure calculation, Python threads cannot run on more than one core at once, and adding threads buys you nothing. The traditional answer has been to use multiple *processes* instead, each with its own interpreter and its own GIL, communicating through channels the operating system provides. This is the subject of Chapter 21.

After three decades of debate, this is now changing. A 2023 proposal, PEP 703, laid out a way to make CPython work correctly without the GIL, and recent versions ship an experimental "free-threaded" build that removes it. Making reference counting safe without a single global lock is delicate work, and the free-threaded build is still maturing, but the long era in which the GIL was simply a fact of Python life is drawing toward its close.

→ **FURTHER READING: THE GIL**

- What Is the Python GIL? *Real Python, the clearest introduction*
- PEP 703: Making the GIL Optional *the free-threading proposal*
- "Understanding the Python GIL" by David Beazley *the famous deep-dive talk*

2.8 How Python Evolves

A living language must change without fracturing. Python's mechanism for this is the **Python Enhancement Proposal**, or PEP. A PEP is a design document proposing a new feature, a change to the language, or a process. It is written up, discussed in public, revised, and either accepted or rejected. Every significant feature of modern Python, from f-strings to type hints to the removal of the GIL, began as a PEP and can be read in full today. They are numbered, and a handful are worth knowing by heart.

PEP	Subject
PEP 1	How the PEP process itself works
PEP 8	The style guide: how to format Python code
PEP 20	The Zen of Python
PEP 484	Type hints
PEP 572	The walrus operator, `:=`
PEP 703	Making the GIL optional

For its first twenty-seven years, final decisions rested with Guido van Rossum, who held the half-joking title of *Benevolent Dictator For Life*. In 2018, after an unusually bitter argument over PEP 572, the walrus operator, Guido stepped down from that role. Governance now rests with an elected **Steering Council** of five core developers, chosen by the people who maintain the language. The arrangement, defined in PEP 13, has held steady ever since, and Python's evolution has continued without a single benevolent dictator to settle disputes.

> **A HABIT WORTH FORMING** When you encounter a Python feature you do not understand, search for "PEP" and the feature's name. The PEP will usually tell you not just what the feature does but *why* it was added, what alternatives were rejected, and what problems it was meant to solve. Few languages document their own reasoning so openly.

→ FURTHER READING: GOVERNANCE & EVOLUTION

- The PEP Index *every proposal, searchable*
- PEP 13: Python's Governance *how the Steering Council works*
- discuss.python.org *where the language's future is debated in the open*

§ § §

Exercises

EXERCISE 2.1 Specification versus implementation

In your own words, explain the difference between "Python the language" and "CPython the implementation". Name one thing an alternative implementation such as PyPy is free to change, and one thing it must keep the same.

EXERCISE 2.2 Read the Zen

Run `import this` in a Python interpreter. Pick the three aphorisms that mean the least to you right now. Write one sentence for each guessing what it might mean. Keep your answers; revisit them after Chapter 19.

EXERCISE 2.3 Disassemble a function

Write a function that takes two numbers and returns their product plus one. Use `dis.dis` to print its bytecode. Identify which instruction performs the multiplication and which performs the addition. How many instructions does the whole function compile to?

EXERCISE 2.4 Watch a reference count

Using `sys.getrefcount(obj)`, observe how the reference count of a list changes as you bind it to a second name and then delete that name. Note: `getrefcount` itself temporarily adds one reference, so the number is always one higher than you might expect. Explain why.

EXERCISE 2.5 Reason about the GIL

You have two programs. Program A downloads 100 web pages. Program B computes the prime factors of 100 large numbers. For each, decide whether Python threads would speed it up on an 8-core machine, and explain your reasoning in terms of the GIL.

EXERCISE 2.6 Trace the pipeline

Describe what happens, stage by stage, when you run `python hello.py`, from the moment you press Enter to the moment output appears. Use the vocabulary of Section 2.5: tokenizer, parser, AST, compiler, bytecode, virtual machine.

EXERCISE 2.7 Find a PEP

The walrus operator `:=` was added by PEP 572 and was controversial enough to prompt Guido's resignation as BDFL. Read the "Rationale" section of PEP 572. Summarise in three sentences what problem it was meant to solve and one objection people raised against it.

EXERCISE 2.8 A research question

PyPy runs the same Python code as CPython but is often several times faster. Spend fifteen minutes researching how a just-in-time compiler achieves this. Write a paragraph explaining, at a high level, why running code can sometimes be faster than interpreting it line by line, and what CPython gives up by not doing so.

Solutions to Chapter 2 Exercises

Solution 2.1: Specification versus implementation

"Python the language" is the set of rules describing what programs mean, written in the Language Reference. "CPython the implementation" is an actual program, written in C, that executes those rules. An implementation like PyPy is free to change *how* it runs code: it may compile hot loops to machine code, use a different garbage collector, or lay out objects in memory differently. It must keep the same *observable behaviour*: `2 + 2` must still be `4`, a `for` loop must still iterate, and a correct Python program must produce the same results. Implementations compete on speed, memory, and platform, not on meaning.

Solution 2.2: Read the Zen

Answers will vary. Two that often puzzle beginners: "There should be one obvious way to do it" is a reaction against languages that offer five interchangeable ways to write the same loop, which fragments how code is read across a team. "Although that way may not be obvious at first unless you're Dutch" is simply a joke about Guido. The value of the exercise is in returning to your guesses later and seeing how your understanding has grown.

Solution 2.3: Disassemble a function

PYTHON

```
import dis
def f(a, b):
    return a * b + 1
dis.dis(f)
```

OUTPUT

```
  2     RESUME              0
  3     LOAD_FAST           a
        LOAD_FAST           b
        BINARY_OP           5 (*)
        LOAD_CONST          1 (1)
        BINARY_OP           0 (+)
        RETURN_VALUE
```

The `BINARY_OP 5 (*)` performs the multiplication; `BINARY_OP 0 (+)` performs the addition. The function compiles to seven instructions (the exact count and form vary slightly between Python versions, which is itself a useful lesson: bytecode is an implementation detail, not part of the language specification).

Solution 2.4: Watch a reference count

```python
import sys
a = [1, 2, 3]
print(sys.getrefcount(a))   # 2: the name a, plus getrefcount's own argument
b = a
print(sys.getrefcount(a))   # 3: a, b, and the argument
del b
print(sys.getrefcount(a))   # 2 again
```

When you call `getrefcount(a)`, the value `a` is passed as an argument to that function, which means that during the call there is one extra reference to the object: the function's parameter. So the reported count is always one higher than the number of references in your own code. This is a small but perfect illustration of the principle that, in Python, passing a value to a function creates another reference to it.

Solution 2.5: Reason about the GIL

Program A (downloading 100 web pages) is input-output bound: most of its time is spent waiting for the network. A waiting thread releases the GIL, so other threads can run while it waits. Threads *would* speed this up significantly. Program B (factoring large numbers) is CPU bound: it is pure computation with no waiting. Because the GIL allows only one thread to execute Python bytecode at a time, the threads would take turns on a single core and the 8 cores would sit mostly idle. Threads would *not* help; you would use multiple processes (or a free-threaded build) instead.

Solution 2.6: Trace the pipeline

When you press Enter, the operating system starts a CPython process and hands it `hello.py`. The *tokenizer* reads the file's characters and groups them into tokens. The *parser* arranges those tokens into an Abstract Syntax Tree representing the program's structure; if the syntax is invalid, you get a `SyntaxError` here, before anything runs. The *compiler* walks the tree and emits bytecode, which may be cached as a `.pyc` file. The *virtual ma-*

chine then executes the bytecode one instruction at a time, and when it reaches a `print` call, output appears on your screen through the operating system's input-output services.

Solution 2.7: Find a PEP

PEP 572 added assignment expressions, written `:=`, which let you bind a value to a name in the middle of a larger expression, for instance inside a `while` condition or a comprehension, so the value need not be computed twice or assigned on a separate line. The chief objection was that it introduced a second way to assign (alongside `=`) and could be used to write dense, hard-to-read one-liners, in tension with the Zen's preference for one obvious way and for readability. The argument grew heated enough that Guido, exhausted by it, resigned as BDFL shortly afterward.

Solution 2.8: A research question

A just-in-time (JIT) compiler watches a program as it runs, identifies the "hot" code that executes most often (typically loops), and compiles that code to native machine instructions on the fly, so subsequent runs of that code skip interpretation entirely and run at near-C speed. It can also specialise the machine code to the actual types it observes at runtime. CPython interprets each bytecode instruction every time, paying the interpretation cost on every pass through a loop. By not having a JIT, CPython stays far simpler: easier to maintain, easier to embed in other programs, and far easier to extend with C libraries, which is the trade its designers chose. (Recent CPython versions have begun adding a limited JIT of their own, narrowing the gap.)

CHAPTER 3 · PART I · THE GROUND

The Object Model

Here Python truly begins. Before the first number or string, there is one idea from which the entire language unfolds: everything you can name is an object, and every object has the same three properties. Grasp this chapter and the rest of the book becomes a series of corollaries.

> *Everything is an object. The big idea is messaging. The key in making great and growable systems is much more to design how its modules communicate rather than what their internal properties and behaviors should be.*
>
> — Alan Kay, who coined the term "object-oriented programming"

Most introductions to Python begin with numbers and strings, treating them as different kinds of thing with different rules, and only much later, if at all, reveal that they were the same kind of thing all along. We are going to do the opposite. We will spend this chapter on the single idea that unifies the whole language, and only then, in the chapters that follow, look at the particular objects it gives you. The order is unusual, but it is the order that makes everything afterward simple. When you understand that a number, a string, a list, a function, and a class are all objects obeying one set of rules, you stop memorising special cases and start reasoning from principle.

The promise of this chapter is concrete. By its end you will understand exactly what a variable is (not what most beginners think), why two equal-looking values are sometimes the same object and sometimes not, why changing a list inside a function changes it outside but changing a number does not, why a list cannot be a dictionary key, and how the `+` sign knows how to add two numbers and also join two strings. These are not five facts to memorise. They are five consequences of one idea.

3.1 Everything Is an Object

In some languages, the world is divided in two. There are *primitives*, the small fast things like integers and characters that the hardware understands directly, and there are *objects*, the larger structured things you build yourself. The two obey different rules. A primitive cannot have methods; an object can. This division is a constant low tax on the programmer's attention, who must always remember which kind of thing they are holding.



Figure 3.1. *Alan Kay, who coined "object-oriented programming" at Xerox PARC in the 1970s.* Alan Kay

Python abolishes the division. In Python, **everything is an object**: the integer `42`, the string `"hello"`, the list `[1, 2, 3]`, the function you define, the class you write, the module you import, and even the types themselves. Every one of them is an object with the same uniform interface. This is why you can ask an integer for its methods, store a function in a list, pass a class as an argument, and treat all of them with the same handful of built-in functions.

EXAMPLE 3.1 Asking different things the same questions

The built-in `type()` reports any object's type, and `type()` works on everything, because everything is an object:

```python
type(42)            # <class 'int'>
type("hello")       # <class 'str'>
type([1, 2, 3])     # <class 'list'>
type(print)         # <class 'builtin_function_or_method'>
type(int)           # <class 'type'>, even a type is an object
type(type)          # <class 'type'>, and its type is type itself
```

The last two lines deserve a pause. The type of `int` is `type`. The type of `type` is `type`. The hierarchy does not regress forever; it closes on itself. `type` is the object whose job is to make other types, and it is its own type. This is the still point at the centre of the whole model.

A practical consequence is that functions and classes are *first-class*: they can be assigned to names, stored in containers, passed as arguments, and returned from other

functions, exactly like any other value. Much of Python's expressive power, and the whole of Chapter 16 on decorators, rests on this.

→ **FURTHER READING: OBJECTS EVERYWHERE**

- The Python Data Model *the reference chapter this one is based on*
- "Python's Class Development Toolkit" by Raymond Hettinger *objects from the ground up*
- First-class functions *Wikipedia*

3.2 Identity, Type, and Value

Every object in Python, without exception, has exactly three intrinsic properties. Understanding them, and which of them can change, resolves most of the confusions that follow.

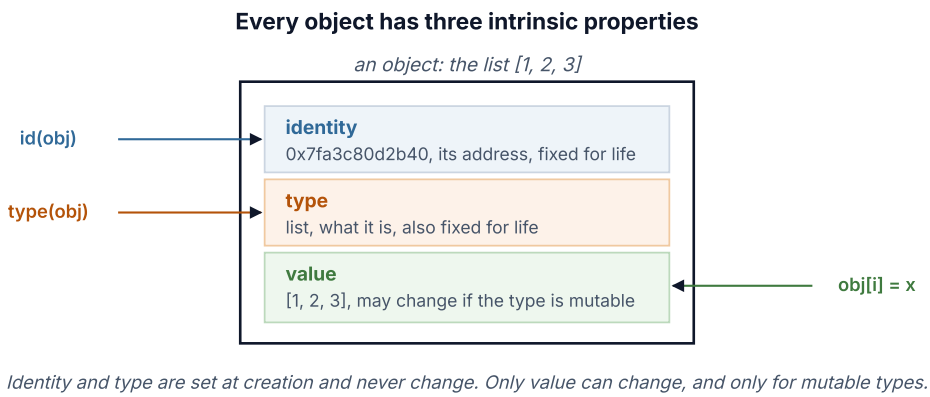


Figure 3.2. The anatomy of any Python object. In CPython, the identity is literally the object's address in memory, which is why it never changes: the object does not move.

The **identity** is what distinguishes an object from every other object that exists at the same time. It is fixed the moment the object is created and never changes for as long as the object lives. You can read it with the built-in `id()`. In CPython, the identity is simply the object's address in memory. The operator `is` compares identities: `a is b` asks whether `a` and `b` are the very same object, not merely equal ones.

The **type** determines what an object is and therefore what it can do: what operations it supports, what methods it has, how it behaves. Like identity, an object's type is fixed at

creation. You read it with `type()`. The type is itself an object, as we saw in Section 3.1.

The **value** is the data the object holds. Here is the one property that may change: whether it can depends entirely on the object's type. Types whose values can change after creation are called *mutable*; types whose values cannot are called *immutable*. This single distinction, the subject of Section 3.4, is the most consequential fact about any Python object.

> **DEF** **Identity**
>
> The property that uniquely distinguishes an object during its lifetime, returned by `id(obj)` and compared by the `is` operator. Two objects with identical values may still have different identities. In CPython, identity is the object's memory address. Do not rely on the specific number; rely only on whether two identities are equal.

→ **FURTHER READING: IDENTITY, TYPE, VALUE**

- Objects, values and types *the Python reference, Section 3.1*
- Python "is" vs "==": Identity vs Equality *Real Python*

3.3 Names Are Not Boxes

We now reach the single most important correction this book can offer a beginner. In school mathematics, and in many programming languages, a variable is a box: `x = 5` opens a box labelled `x` and puts a `5` inside it. Assigning `x = 7` replaces the contents of the box. This mental model is wrong for Python, and the difference is not academic; it is the source of more genuine confusion than any other feature of the language.

In Python, a variable is not a box. It is a **name**, and a name is a label that is *bound* to an object that exists elsewhere. The statement `x = 5` does three things, in order: it creates (or finds) an integer object with the value 5, it creates the name `x`, and it binds the name to the object, like tying a luggage tag to a suitcase. The name is not the object. The name is a way to reach the object.

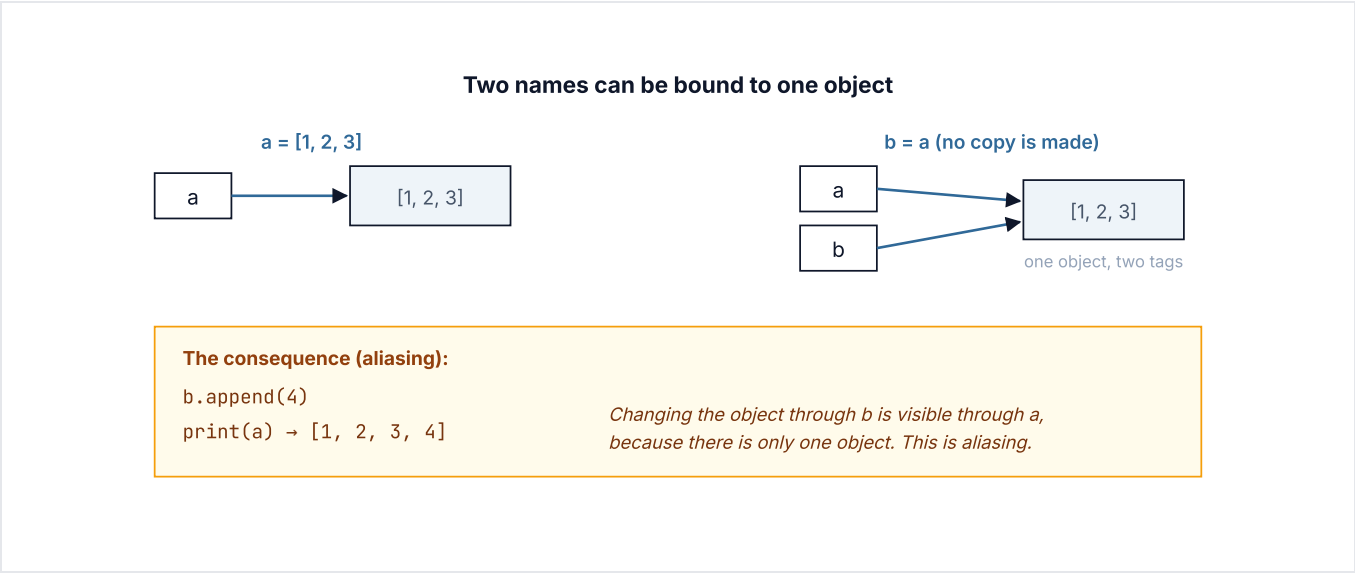


Figure 3.3. Binding, not copying. The statement `b = a` binds a second name to the same object; it does not duplicate it. This is the same shared object whose reference count we watched in Figure 2.5.

Because a name is only a label, several names can be bound to one object at once. This is called **aliasing**, and it is the source of a beginner's first truly baffling bug. When you write `b = a`, you do not copy the object; you bind a second name to it. If the object is mutable and you change it through one name, the change is visible through the other, because there was only ever one object.

EXAMPLE 3.2 Aliasing, and how to avoid it

The shared-object behaviour, and the three usual ways to ask for a genuine copy:

```python
a = [1, 2, 3]
b = a                # b is bound to the SAME list
b.append(4)
print(a)             # [1, 2, 3, 4], a sees it too
print(a is b)        # True, one object, two names

# To get an independent copy, ask for one explicitly:
c = a.copy()         # or  a[:]  or  list(a)
c.append(5)
print(a)             # [1, 2, 3, 4], unchanged
print(a is c)        # False, different objects
```

A shallow copy duplicates the outer list but not the objects inside it. If your list contains other mutable objects (lists of lists), you need `copy.deepcopy()` to copy all the way down. We return to this in Chapter 8.

Reassignment fits the same model. When you write `x = 7` after `x = 5`, you do not change the integer 5 into a 7. You bind the name `x` to a different object, the integer 7. The 5 is untouched; if nothing else refers to it, its reference count drops to zero and it is reclaimed, exactly as Chapter 2 described.

→ **FURTHER READING: NAMES AND BINDING**

- "Facts and Myths about Python names and values" by Ned Batchelder *the definitive explanation, with diagrams*
- Ned Batchelder's PyCon talk on the same *video version*

3.4 Mutability

An object is **mutable** if its value can change after it is created, and **immutable** if it cannot. This property belongs to the type, not the individual object: every list is mutable, every string is immutable, and no exceptions exist. The table below sorts the built-in types.

Immutable	Mutable
`int`, `float`, `complex`, `bool`	`list`
`str`	`dict`
`tuple`	`set`
`frozenset`	`bytearray`
`bytes`	most objects you define yourself
`None`	

The crucial subtlety is what it means to "change" an immutable object: you cannot, and any operation that appears to is in fact creating a new object and rebinding your name to it. When you write `s = s.upper()`, the string method `upper()` does not alter the original string; it builds and returns a new one, and the assignment binds `s` to that new string. The original is unchanged, and if no other name holds it, it is reclaimed.

EXAMPLE 3.3 Immutable means a new object every time

PYTHON

```python
s = "hello"
print(id(s))        # e.g. 140234881234   (some address)
s = s.upper()
print(s)            # HELLO
print(id(s))        # DIFFERENT address, a new object

# Strings reject in-place change outright:
s[0] = "J"          # TypeError: 'str' object does not support item assignmen

# A list, being mutable, accepts it:
nums = [1, 2, 3]
before = id(nums)
nums[0] = 99
print(id(nums) == before)   # True, same object, changed in place
```

THE CLASSIC TRAP Because mutability is a property of the type, a mutable object used as a function's default argument is created *once*, when the function is defined, and then shared across every call. This produces one of the most surprising bugs in the language. We cover it in full in Chapter 15, but it is, at root, simply this section's principle applied to a default value.

→ **FURTHER READING: MUTABILITY**

- Mutable and immutable objects *Python reference*
- Python's Mutable vs Immutable Types *Real Python, with memory diagrams*

3.5 What Happens When You Pass an Object to a Function

Programmers arriving from other languages often ask whether Python passes arguments "by value" or "by reference". The honest answer is that it does neither, in the sense those terms usually carry, and the confusion dissolves the moment you apply the model of names and objects you already have. Python passes arguments by *binding the parameter name to the same object the caller passed*. The name inside the function and the name outside it are two labels on one object. Some call this "call by object reference" or "call by sharing".

From this single rule, the behaviour that surprises people follows necessarily. If the object is mutable and the function mutates it in place, the caller sees the change, because there is one object. If the function instead rebinds its parameter to a new object, the caller sees nothing, because rebinding a local name has no effect on the caller's name.

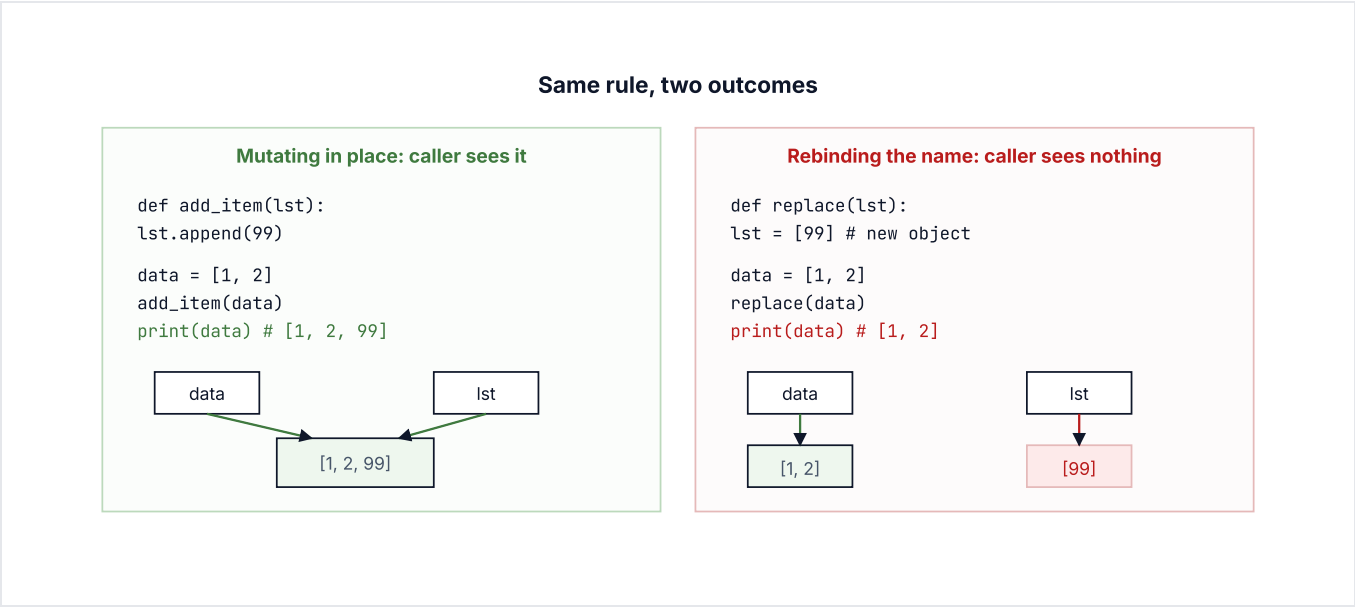


Figure 3.4. Both functions receive the same object. The left mutates it, so the caller's list changes. The right rebinds its local name to a brand-new list, leaving the caller's binding untouched.

There is nothing special about functions here. The parameter is simply a new name, bound by the call to the argument object, in exactly the way `b = a` binds a new name in ordinary code. Once you see function calls as binding, you never again have to memorise a separate rule for them.

→ **FURTHER READING: ARGUMENT PASSING**

- Python FAQ: call by reference? *the official answer*
- "Python's pass-by-object-reference" *a memorable explanation*

3.6 Equality versus Identity

Two questions can be asked of a pair of objects, and beginners frequently confuse them. *Are these the same object?* is a question of identity, answered by `is`. *Do these objects have the same value?* is a question of equality, answered by `==`. They are different questions with sometimes different answers, and using the wrong operator is a common and subtle bug.

PYTHON

```
a = [1, 2, 3]
b = [1, 2, 3]    # a separate list with equal contents
c = a
```

```
a == b      # True, same value
a is b      # False, different objects
a is c      # True, same object
```

The rule of thumb is short. Use `==` to compare values, which is what you want almost all the time. Use `is` only to ask about identity, and in practice that means almost exclusively comparing against `None`, which is a unique object: write `if x is None`, never `if x == None`. The reason `is None` is preferred is both speed and correctness, since a perverse class could define `==` to lie, but nothing can fake identity with the one true `None`.

One quirk of CPython tempts beginners into misusing `is`, and it is worth seeing so that you are never fooled by it. For efficiency, CPython keeps a single cached copy of every small integer, from -5 to 256, and reuses it. So two names set to `100` happen to point at the same cached object, and `is` returns `True`. But two names set to `1000` get separate objects, and `is` returns `False`. The values are equal in both cases; the identities are an implementation detail you must never depend on.

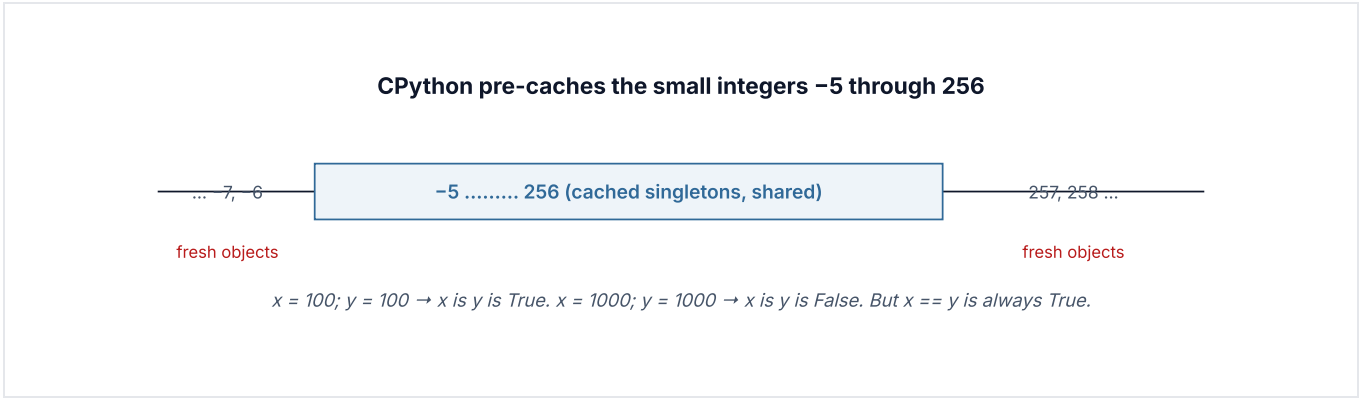


Figure 3.5. The small-integer cache. This is the canonical example of why `is` must never be used to compare values: the answer depends on an optimisation, not on the program's meaning.

→ **FURTHER READING: == VS IS**

- Comparisons *Python reference*
- "Is there a difference between == and is?" *the canonical Stack Overflow answer*

3.7 Hashability

Some objects can serve as dictionary keys or set members, and some cannot. The dividing property is called **hashability**. An object is hashable if it has a hash value that never

changes during its lifetime, provided by a method called `__hash__`, and if it can be compared for equality with other objects. Dictionaries and sets use this hash value to find things quickly, a mechanism we explain fully in Chapter 10.

The practical rule connects directly to mutability. Immutable built-in objects, numbers, strings, tuples (of hashable things), and frozensets, are hashable and may be used as keys. Mutable built-in objects, lists, dictionaries, and sets, are not hashable and may not. The reason is a matter of necessity, not arbitrary rule: if you could change an object after using it as a key, its hash would change, and the dictionary would lose track of where it stored the entry.

```python
d = {}
d[(1, 2)] = "a point"      # tuple key: fine, tuples are hashable
d["name"] = "Ada"          # str key: fine

d[[1, 2]] = "oops"         # TypeError: unhashable type: 'list'

hash((1, 2))               # a fixed integer, e.g. -3550055125485641917
hash([1, 2])               # TypeError: unhashable type: 'list'
```

THE CONTRACT Hashability ties together equality and hashing by one rule: if two objects are equal, they must have the same hash. Objects you design yourself inherit a sensible default, but if you override `__eq__` to define your own notion of equality, you must override `__hash__` to match, or your objects will misbehave as dictionary keys. Chapter 19 covers how to do this correctly.

→ **FURTHER READING: HASHING**

- Glossary: hashable *Python docs*
- object.__hash__ *the hash/eq contract in full*

3.8 The Data Model: How Objects Speak a Common Language

We can now answer the question posed at the start of the chapter: how does the `+` sign know how to add two numbers and also join two strings and also concatenate two lists?

The answer reveals the machinery beneath the whole language. Python defines a set of **special methods**, recognisable by their double-underscore names and universally nicknamed *dunder* methods, that objects implement to participate in the language's syntax. When you write `a + b`, Python does not have special knowledge of addition. It calls `a.__add__(b)`, and the object decides what addition means for its type.

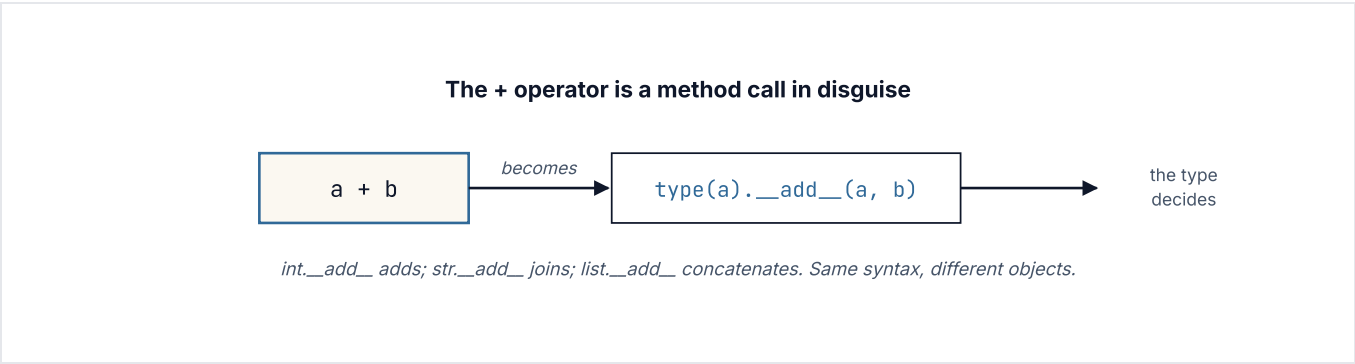


Figure 3.6. Operator dispatch. Nearly every piece of Python syntax, arithmetic, indexing, iteration, calling, comparison, is a disguised call to a dunder method, which is why your own classes can participate in all of it.

This is the deepest reason "everything is an object" is more than a slogan. The syntax of the language is uniform precisely because it is all implemented as method calls on objects, and an object of any type, including one you write yourself, can implement those methods and thereby behave like a built-in. A table of the most common protocols:

You write	Python calls	Protocol
`a + b`	`a.__add__(b)`	arithmetic
`a == b`	`a.__eq__(b)`	comparison
`len(a)`	`a.__len__()`	sizing
`a[i]`	`a.__getitem__(i)`	indexing
`for x in a`	`a.__iter__()`	iteration
`a(x)`	`a.__call__(x)`	calling
`print(a)`	`a.__str__()`	display
`repr(a)`	`a.__repr__()`	debugging display
`with a:`	`a.__enter__()` / `__exit__()`	context management

EXAMPLE 3.4 A type that behaves like a built-in

By implementing the right dunder methods, a class you write supports the same syntax as any built-in. Here a two-dimensional vector that can be added with `+` and printed legibly:

PYTHON

```python
class Vector:
    def __init__(self, x, y):
        self.x, self.y = x, y

    def __add__(self, other):
        return Vector(self.x + other.x, self.y + other.y)

    def __repr__(self):
        return f"Vector({self.x}, {self.y})"

v = Vector(1, 2) + Vector(3, 4)
print(v)            # Vector(4, 6)
```

The `+` worked on a type that did not exist until you wrote it, because `+` was never about numbers. It was always about `__add__`. This is the whole of Chapter 19 in miniature.

→ **FURTHER READING: THE DATA MODEL**

- Special method names *the complete list*
- Luciano Ramalho, *Fluent Python* *the definitive book on the data model*
- A Guide to Python's Magic Methods *free, well organised*

3.9 Truthiness

One more protocol deserves its own section, because it shapes how you write conditions throughout the language. Every object in Python can be tested for truth, in an `if` or a `while` or with the `and`, `or`, and `not` operators, not only the values `True` and `False`. When asked, an object reports its own truthiness through the `__bool__` method, or failing that, through `__len__` (an object with length zero is false). The values that count as false are few and worth memorising.

Falsy (count as False)	Truthy (count as True)
`False`, `None`	everything not in the falsy column
`0`, `0.0`, `0j`, `Decimal(0)`	any non-zero number
`""` (empty string)	any non-empty string
`[]`, `()`, `{}`, `set()`	any non-empty container

The idiomatic consequence is that you test a container's emptiness directly, rather than comparing its length to zero. Write `if not names:` rather than `if len(names) == 0:`; the first is shorter, reads more naturally, and works for any object that defines truthiness. This habit, small in itself, is one of the clearest marks of code written by someone who understands the object model rather than merely uses it.

EXAMPLE 3.5 Truthiness in practice

PYTHON

```python
names = []
if not names:
    print("the list is empty")      # this runs

# A custom object decides its own truth:
class ShoppingCart:
    def __init__(self):
        self.items = []
    def __bool__(self):
        return len(self.items) > 0

cart = ShoppingCart()
if not cart:
    print("your cart is empty")     # this runs
```

→ **FURTHER READING: TRUTHINESS**

- Truth Value Testing *Python docs*
- object.__bool__ *how to define your own*

§ § §

Exercises

EXERCISE 3.1 Everything has a type

Predict the output of `type(type([]))` before running it. Then run it. Explain in one sentence why the answer is what it is, referring to Section 3.1.

EXERCISE 3.2 Predict the aliasing

Without running it, predict what each `print` shows. Then run it to check.

```python
a = [1, 2, 3]
b = a
c = a[:]
b.append(4)
c.append(5)
print(a)
print(b)
print(c)
```

EXERCISE 3.3 Immutable rebinding

Write code that demonstrates, using `id()`, that `x += 1` on an integer produces a new object, while `lst.append(1)` on a list keeps the same object. Explain the difference in terms of mutability.

EXERCISE 3.4 The function-call puzzle

Write two functions, each taking one list argument. The first should mutate the list so the change is visible to the caller. The second should attempt to replace the list but leave the caller's list unchanged. Demonstrate both, and explain why they differ using the model from Section 3.5.

EXERCISE 3.5 is versus ==

Find a pair of values for which `==` is `True` but `is` is `False`, and a pair for which both are `True`. Then explain why `x = 256; y = 256; x is y` behaves differently from `x = 257; y = 257; x is y`.

EXERCISE 3.6 Why can't a list be a key?

Try to use a list as a dictionary key and observe the error. Then explain, in terms of the relationship between mutability and hashing, why allowing it would break the dictionary. What hashable object would you use instead to represent the same data?

EXERCISE 3.7 Implement a dunder

Extend the `Vector` class from Example 3.4 with a `__eq__` method so that `Vector(1, 2) == Vector(1, 2)` is `True`. Then check whether your vectors can be added to a set. What happens, and what additional method would you need to fix it?

EXERCISE 3.8 Truthiness

For each of these values, predict whether `bool(value)` is `True` or `False`: `0`, `0.0`, `""`, `" "`, `[]`, `[0]`, `None`, `{}`, `{0: 0}`. Run `bool()` on each to confirm. Which one most often surprises beginners, and why?

Solutions to Chapter 3 Exercises

Solution 3.1: Everything has a type

`type([])` is `<class 'list'>`, and `type(list)` is `<class 'type'>`, so `type(type([]))` is `<class 'type'>`. The answer is `type` because every type is itself an object whose type is `type`, the object responsible for creating types. The chain stops here because `type` is its own type.

Solution 3.2: Predict the aliasing

`a` prints `[1, 2, 3, 4]`, `b` prints `[1, 2, 3, 4]`, and `c` prints `[1, 2, 3, 5]`. Because `b = a` binds `b` to the same object as `a`, appending through `b` is visible through `a` (they are one list). But `c = a[:]` makes an independent copy, so appending 5 to `c` affects only `c`. The lesson: `=` shares, slicing copies.

Solution 3.3: Immutable rebinding

```python
x = 10
print(id(x)); x += 1; print(id(x))   # two DIFFERENT ids

lst = [1]
print(id(lst)); lst.append(2); print(id(lst))   # SAME id
```

An `int` is immutable, so `x += 1` cannot change the existing integer; it computes a new integer and rebinds `x` to it, giving a new `id`. A `list` is mutable, so `append` changes the existing object in place, and its `id` is unchanged.

Solution 3.4: The function-call puzzle

```python
def mutate(lst):
    lst.append(99)        # changes the shared object

def rebind(lst):
    lst = [99]            # binds the LOCAL name to a new object

data = [1, 2]
mutate(data); print(data)   # [1, 2, 99]
rebind(data); print(data)   # [1, 2, 99], rebind did nothing visible
```

`mutate` changes the one object both names point at, so the caller sees it. `rebind` only re-points its local parameter at a new list; the caller's name still points at the original, so nothing changes for the caller. The parameter is just another name bound by the call.

Solution 3.5: `is` versus `==`

Two distinct lists with equal contents (`[1] == [1]` is `True`, `[1] is [1]` is `False`) satisfy the first case; any object compared with itself, or `None is None`, satisfies the second. The integers `256` are the same object because CPython caches every small integer from -5 to 256 and reuses it, so `is` is `True`. The integer `257` is outside that cache, so

each occurrence is a fresh object and `is` is `False`. In both cases `==` is `True`, which is why you must use `==` for value comparison.

Solution 3.6: Why can't a list be a key?

Using a list as a key raises `TypeError: unhashable type: 'list'`. A dictionary stores each entry in a location derived from its key's hash. If the key were mutable, you could change it after insertion, its hash would change, and the dictionary would look in the wrong place and fail to find the entry. Forbidding mutable keys prevents this corruption. To represent the same ordered data as a key, use a `tuple`, which is immutable and therefore hashable: `d[(1, 2)] = ...`.

Solution 3.7: Implement a dunder

```python
class Vector:
    def __init__(self, x, y):
        self.x, self.y = x, y
    def __eq__(self, other):
        return (self.x, self.y) == (other.x, other.y)
```

With `__eq__` defined, `Vector(1, 2) == Vector(1, 2)` is `True`. But adding the vectors to a set raises `TypeError: unhashable type: 'Vector'`. This is by design: when you define `__eq__`, Python sets `__hash__` to `None`, because the default hash (based on identity) would no longer agree with your new equality. To make the vectors usable in a set or as keys, add a matching `__hash__`, for example `def __hash__(self): return hash((self.x, self.y))`, honouring the contract that equal objects share a hash.

Solution 3.8: Truthiness

False: `0`, `0.0`, `""`, `[]`, `None`, `{}`. True: `" "` (a space is a non-empty string), `[0]` (a non-empty list, even though its only element is falsy), `{0: 0}` (a non-empty dict). The one that most often surprises beginners is `[0]`: a list containing a single zero is truthy, because truthiness of a container depends on whether it is *empty*, not on the truthiness of its contents. Likewise `" "` catches people who forget a space is a real character.

CHAPTER 4 · PART II · THE PRIMITIVES

Numbers in Five Forms

Most languages give you two kinds of number and leave you to suffer their limitations. Python gives you five, each a deliberate trade between range, exactness, and speed. Choosing the right one is a real skill, and choosing the wrong one has, more than once, destroyed rockets and bankrupted rounding errors into fortunes.

> *The fast drives out the slow even if the fast is wrong.*
>
> — William Kahan, principal architect of the IEEE 754 floating-point standard

Numbers feel like the simplest thing a computer could possibly handle, and that feeling is a trap. We learned numbers as children and stopped questioning them, but the machine does not have the luxury of the infinite paper on which we were taught to do long division. Every numeric type a computer offers is a compromise: how large a value may grow, how precisely it may be held, how quickly it may be calculated. A language's choice of numeric types is therefore a choice about which compromises to expose and which to hide, and Python's choice is unusually generous. Where C offers a handful of fixed-size integers and two sizes of float, Python offers five distinct numeric types, and understanding when each is the right tool is part of what separates a fluent programmer from a fluent typist.

This chapter covers all five: the unbounded integer, the floating-point real, the exact decimal, the exact rational, and the complex number. It explains how each is stored, where each is exact and where each lies, and how Python decides what happens when you mix them. Everything here is a direct application of Chapter 3: numbers are objects, arithmetic is method dispatch, and the differences between the types are differences in what those objects promise.

4.1 The Numeric Tower

Before the individual types, the structure that relates them. Python organises numbers into an abstract hierarchy, defined formally in the `numbers` module and known informally as the **numeric tower**. Each level is a more specialised kind of number than the one above it, and each concrete type you actually use slots in at the appropriate level.

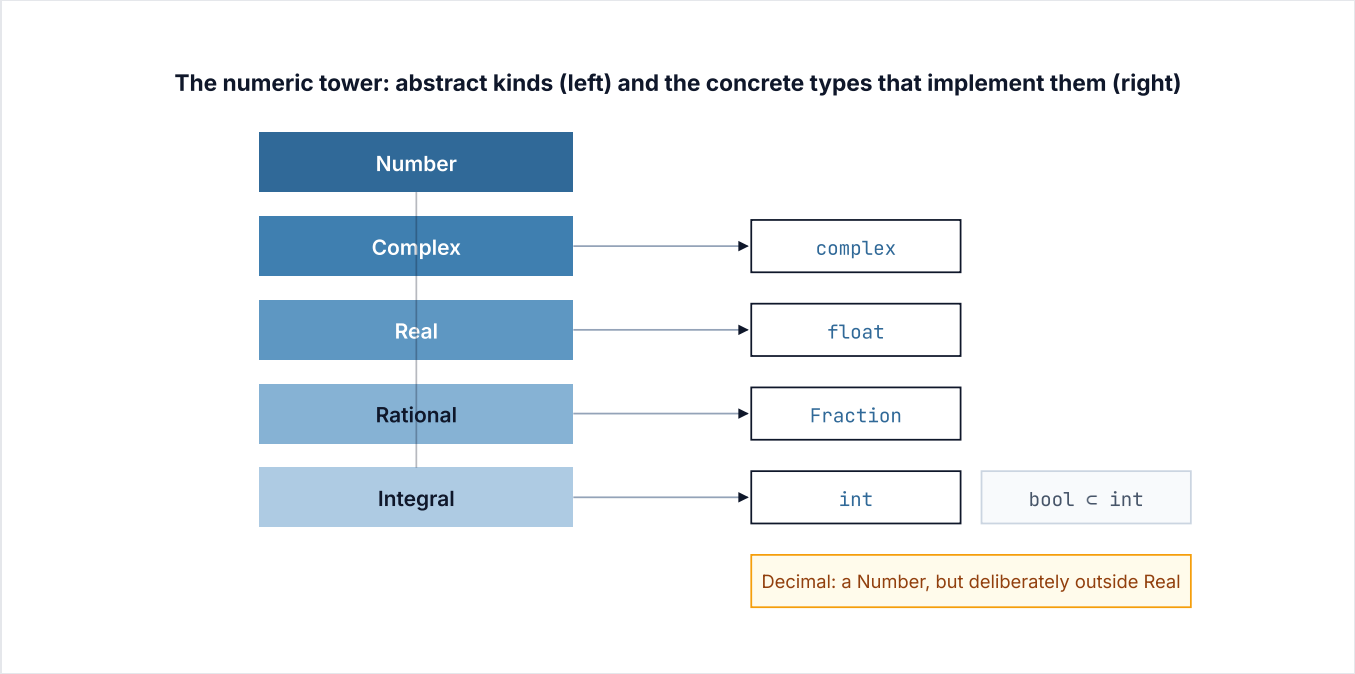


Figure 4.1. The numeric tower. The abstract kinds let code ask general questions (`isinstance(x, numbers.Integral)` means "is this any whole number?"). `Decimal` sits apart on purpose, so it will not silently mix with binary floats.

The tower exists so that code can reason about numbers in general without caring about their exact type. A function that needs any whole number can test `isinstance(n, numbers.Integral)` and accept both `int` and `bool`; a function that needs any real value can test against `numbers.Real`. You will not use the `numbers` module often, but knowing the tower is there explains why `int` and `float` cooperate so smoothly, and why `Decimal` sometimes refuses to.

→ **FURTHER READING: THE NUMERIC TOWER**

- PEP 3141: A Type Hierarchy for Numbers *the design rationale*
- The numbers module *Python docs*

4.2 Integers: Unbounded by Design

Python's `int` has no maximum value. It grows as large as the memory of your machine allows, and arithmetic on it never overflows. This is unlike C, Java, and most other languages, where an integer is a fixed-size box (commonly 32 or 64 bits) that silently wraps around when it overflows, the flaw that, as Chapter 1 described, destroyed the Ariane 5 rocket. Python pays for safety with speed: a Python integer is a full object, not a raw machine word, and arithmetic on it is slower. For the vast majority of programs the safety is worth far more than the speed.

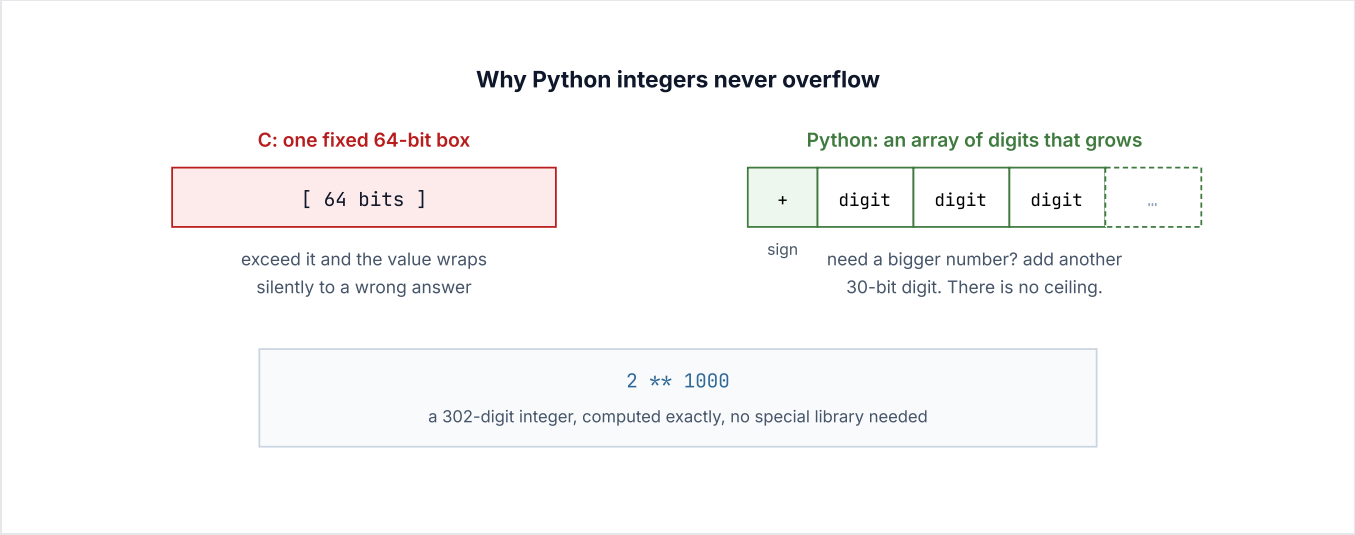


Figure 4.2. CPython stores an integer as a sign plus an array of 30-bit "digits". When a value outgrows the current array, Python allocates a longer one. This is why `2 ** 1000` simply works.

EXAMPLE 4.1 Integers that no fixed-size type could hold

```python
factorial = 1
for i in range(1, 51):
    factorial *= i
print(factorial)
# 30414093201713378043612608166064768844377641568960512000000000000

print((2 ** 1000).bit_length())   # 1001, the number needs 1001 bits
print((255).bit_count())          # 8, number of 1-bits set
```

Literals, bases, and bit operations

Integer literals can be written in four bases, may use underscores as digit separators for readability, and support the full set of bitwise operators inherited from C.

```python
255          # decimal
0b1111_1111  # binary, with an underscore separator
0o377        # octal
0xff         # hexadecimal
int("ff", 16)  # 255, parse a string in a given base

5 & 3     # 1   bitwise AND
5 | 3     # 7   bitwise OR
5 ^ 3     # 6   bitwise XOR
1 << 4    # 16  left shift (multiply by 2**4)
256 >> 2  # 64  right shift (floor-divide by 2**2)
```

→ **FURTHER READING: INTEGERS**

- Numeric types *Python docs*
- How Python implements integers *the digit-array layout in detail*

4.3 Floats and the IEEE 754 Reality

Python's `float` is a 64-bit IEEE 754 double-precision number, the exact layout we dissected in Figure 1.11: one sign bit, eleven exponent bits, fifty-two fraction bits. It is the same floating-point type used by C, Java, JavaScript, and essentially every modern processor, and it carries the same famous hazard. A float is an *approximation* of a real number, not an exact value, and the approximation occasionally surprises you.



Figure 4.3. *William Kahan, who led the IEEE 754 standard and won the 1989 Turing Award for it.* William Kahan

```python
0.1 + 0.2          # 0.30000000000000004
0.1 + 0.2 == 0.3   # False
```

This is not a bug in Python; it is the nature of representing decimal fractions in binary, and Chapter 1 explained why 0.1 has no exact binary form. What is worth adding here is *why*

the interpreter usually hides the problem. Since Python 3.1, the `repr` of a float prints the shortest decimal string that round-trips back to the same underlying value, so `repr(0.1)` shows the tidy `0.1` even though the stored value is the long expansion. The display is honest about the value's identity but merciful about its ugliness. The ugliness only surfaces when arithmetic makes two approximations disagree, as in `0.1 + 0.2`.

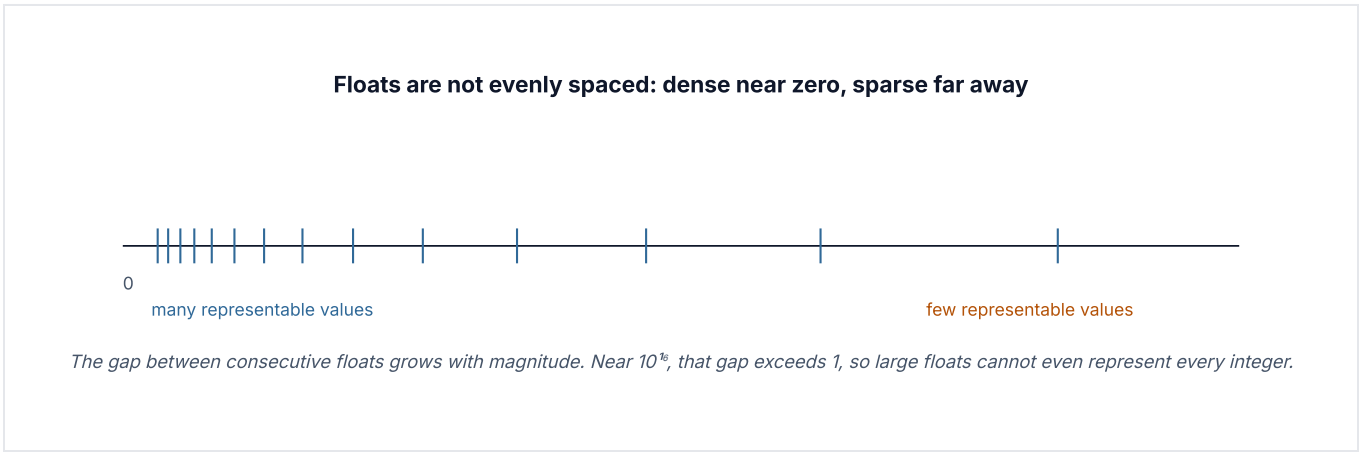


Figure 4.4. Floating-point values are not spread evenly along the number line. This is why precision is lost on large magnitudes, and why `2.0 ** 53 + 1 == 2.0 ** 53` is `True`.

The correct way to compare two floats is therefore not `==` but a tolerance check. The `math` module provides one built for the purpose, `math.isclose`, which compares with both a relative and an absolute tolerance.

EXAMPLE 4.2 Comparing floats safely, and the special values

```python
import math
math.isclose(0.1 + 0.2, 0.3)        # True

# IEEE 754 also defines three values that are not ordinary numbers:
math.inf                             # positive infinity
-math.inf                            # negative infinity
math.nan                             # "not a number", the result of 0/0, i

# nan is the one value not equal to itself:
math.nan == math.nan                 # False!
math.isnan(math.nan)                 # True, the correct way to test

import sys
sys.float_info.max                   # 1.7976931348623157e+308
sys.float_info.epsilon               # 2.220446049250313e-16
```

The behaviour of `nan` is not a Python quirk; the IEEE standard mandates that `nan` compare unequal to everything, including itself. Code that sorts or deduplicates floating-point data must handle it deliberately, or it will behave strangely in ways that are maddening to debug.

HISTORY In 1994, a flaw in the floating-point division unit of Intel's new Pentium processor produced wrong answers for certain divisions. Intel initially dismissed it as rare, then, under public pressure, recalled the chips at a cost of around 475 million dollars. The episode established the modern expectation that arithmetic hardware must be provably correct, and it remains the textbook example of why floating-point behaviour is treated with such seriousness.



Figure 4.5. *An original Intel Pentium. A subtle error in its division lookup table, the "FDIV bug" of 1994, forced a recall costing roughly 475 million dollars and became a lasting lesson in the cost of floating-point mistakes.* Pentium FDIV bug

→ **FURTHER READING: FLOATING POINT**

- The Floating-Point Guide *the friendliest introduction*
- Goldberg, "What Every Computer Scientist Should Know About Floating-Point" *the canonical paper*
- Python tutorial: Floating-Point Arithmetic *official, concise*

4.4 Decimal: When the Cents Must Add Up

When a bank totals an account or a tax authority computes what is owed, floating-point error is not a curiosity but a liability. For these uses Python provides `decimal.Decimal`, a type that performs arithmetic in base ten, exactly as a careful human would with pencil and paper, so that `0.1 + 0.2` is precisely `0.3` and not a hair more.

EXAMPLE 4.3 Decimal arithmetic, and the construction trap

```python
from decimal import Decimal, getcontext, ROUND_HALF_UP

Decimal("0.1") + Decimal("0.2")        # Decimal('0.3'), exact

# THE TRAP: build Decimals from strings, never from floats.
Decimal(0.1)      # Decimal('0.1000000000000000055511151231257...')
Decimal("0.1")    # Decimal('0.1'), clean

# A currency calculation, rounded to cents:
price = Decimal("19.99")
tax   = Decimal("0.0825")
total = (price * (1 + tax)).quantize(Decimal("0.01"), rounding=ROUND_HALF_UP
print(total)        # 21.64
```

The trap is worth stating loudly: `Decimal(0.1)` captures the float's error, because by the time the literal `0.1` reaches the `Decimal` constructor it is already a flawed binary float. `Decimal("0.1")`, from a string, is exact. Always build money values from strings.

A `Decimal` carries a *context* that controls how many significant digits it keeps (the default is 28) and how it rounds (the default is banker's rounding, which we meet again in Section 4.9). You can adjust the context globally with `getcontext()` when a calculation needs more or fewer digits, and `quantize` rounds a value to a fixed number of decimal places, which is exactly what currency requires.

→ **FURTHER READING: DECIMAL**

- The decimal module *Python docs, with a FAQ on money*
- The General Decimal Arithmetic Specification *the standard Python's Decimal implements*

4.5 Fraction: Exact Rationals

Where `Decimal` is exact in base ten, `fractions.Fraction` is exact in every base, because it stores a number as a true ratio of two integers, a numerator and a denominator,

always reduced to lowest terms. One third plus one sixth is exactly one half, with no rounding anywhere, because the arithmetic is done on the integers themselves.

```python
from fractions import Fraction

Fraction(1, 3) + Fraction(1, 6)     # Fraction(1, 2), exact
Fraction("0.25") * 4                 # Fraction(1, 1)
Fraction(6, 8)                       # Fraction(3, 4), auto-reduced

# Recover a clean fraction from a messy float:
Fraction(0.5)                        # Fraction(1, 2)
Fraction(0.1).limit_denominator(1000)  # Fraction(1, 10), the intended value
```

Fractions are the right tool when exactness matters and the values are genuinely rational: probability calculations, symbolic mathematics, and anywhere a result like "two thirds" must stay two thirds rather than becoming `0.6666666666666666`. The `limit_denominator` method is a small gem, recovering the simple fraction a human most likely intended from a float that approximates it.

→ **FURTHER READING: FRACTIONS**

- The fractions module *Python docs*

4.6 Complex Numbers

Python is one of the few mainstream languages with complex numbers built into the core, no import required. A complex literal is written with a `j` suffix on its imaginary part, following the engineering convention rather than the mathematician's `i`. Complex numbers are indispensable in signal processing, electrical engineering, and physics, and having them in the language proper rather than in a library makes that work noticeably cleaner.

```python
z = 2 + 3j
z.real            # 2.0
z.imag            # 3.0
abs(z)            # 3.605551275463989, the magnitude (hypotenuse)
z.conjugate()     # (2-3j)
z * (1 - 2j)      # (8-1j)
```

```
import cmath
cmath.sqrt(-1)      # 1j, the square root math.sqrt cannot give you
cmath.phase(z)    # 0.982..., the angle in radians
```

The companion module `cmath` provides the complex versions of the trigonometric and exponential functions that `math` offers only for reals. Note that `math.sqrt(-1)` raises an error, while `cmath.sqrt(-1)` happily returns `1j`: the two modules embody different promises about what a number is allowed to be.

→ **FURTHER READING: COMPLEX NUMBERS**

- The cmath module *Python docs*

4.7 A Surprise: `bool` Is an `int`

Here is a fact that startles most newcomers and that is, once seen, a small delight. In Python, `bool` is not a separate type alongside `int`; it is a *subclass* of `int`. `True` is, for every arithmetic purpose, equal to `1`, and `False` is equal to `0`. This was done in 2003 for backward compatibility, when booleans were added to a language that had long used 1 and 0 in their place, and it has proven quietly useful ever since.

PYTHON

```
True == 1                 # True
False == 0                # True
isinstance(True, int)     # True
True + True               # 2

# The practical payoff: counting with sum()
votes = [True, False, True, True, False]
sum(votes)                # 3, counts the True values

scores = [82, 45, 91, 67, 30]
sum(s >= 60 for s in scores)  # 3, how many passed
```

That last idiom, summing a sequence of comparisons to count how many are true, is genuinely Pythonic and reads cleanly once you know that a boolean is a number. It is a direct

consequence of the object model: `True` is an object whose type happens to be a subclass of `int`, so it inherits every numeric behaviour.

→ **FURTHER READING: BOOLEANS AS INTEGERS**

○ PEP 285: Adding a bool type *why bool subclasses int*

4.8 Mixing Types: Operator Dispatch and Division

What happens when you add an `int` to a `float`? The answer applies the data model from Chapter 3. The expression `1 + 2.0` first asks the integer, `(1).__add__(2.0)`. The integer does not know how to add a float to itself and returns the special value `NotImplemented`. Python then asks the float, from the other side, `(2.0).__radd__(1)`, and the float, which does know how to absorb an integer, converts the `1` to `1.0` and returns `3.0`. The result climbs the numeric tower to the more general type.

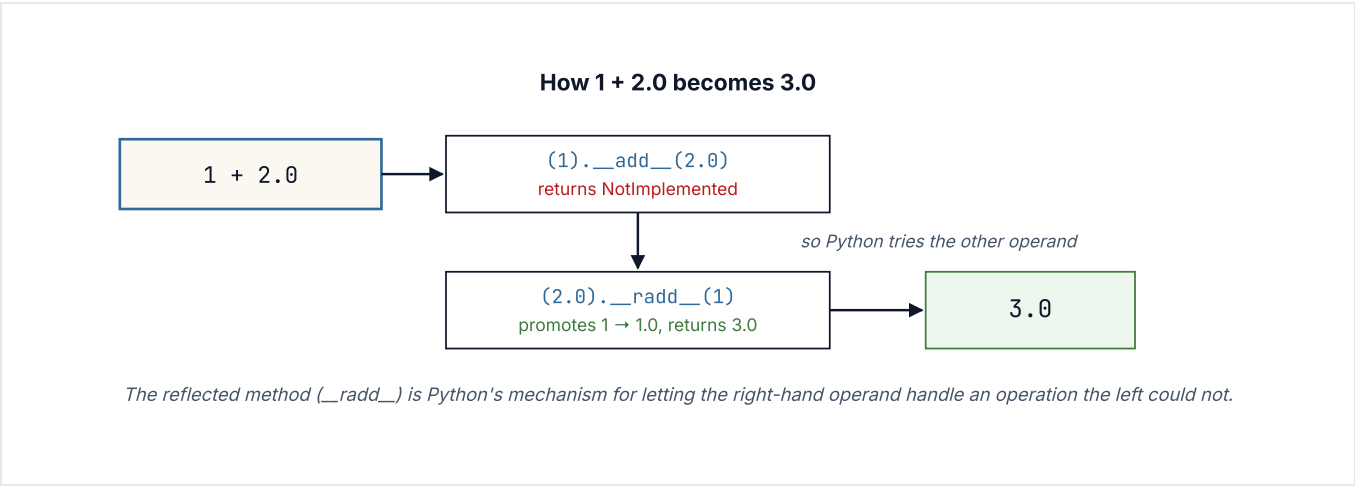


Figure 4.6. Mixed-type arithmetic resolved by reflected operators. This is the same dispatch shown in Figure 3.6, now with the fallback step that lets the second operand take over.

The two divisions, and the sign of the remainder

Python has two division operators, and the difference trips up nearly everyone once. True division, `/`, always produces a float, even when the operands divide evenly: `6 / 2` is `3.0`, not `3`. Floor division, `//`, rounds the result down toward negative infinity, and the modulo operator, `%`, returns a remainder whose sign follows the *divisor*, not the dividend. This last rule differs from C and surprises people, but it has the elegant property

that `a % n` is always in the range from 0 up to `n` for a positive `n`, which is usually what you want.

```python
7 / 2        # 3.5 , always a float
7 // 2       # 3   , floor division
-7 // 2      # -4  , rounds toward negative infinity, not toward zero
-7 % 2       # 1   , sign follows the divisor (positive here)
divmod(17, 5)  # (3, 2), quotient and remainder together
```

→ **FURTHER READING: OPERATORS**

- Emulating numeric types *__add__, __radd__, and the rest*
- Guido: "Why Python's integer division floors" *the rationale, first-hand*

4.9 The math Toolkit, and One Last Trap

Python's numeric work is supported by a family of standard-library modules. You will reach for these constantly, and knowing which one owns which job saves a great deal of searching.

Module	Use it for
`math`	real-valued functions: `sqrt`, `log`, `sin`, `floor`, `gcd`, `isclose`, `factorial`
`cmath`	the same, for complex numbers
`decimal`	exact base-ten arithmetic (money)
`fractions`	exact rational arithmetic
`statistics`	`mean`, `median`, `stdev`, `quantiles`
`random`	random numbers, choices, shuffling, sampling

A few `math` functions are worth knowing by name because they solve problems people often solve worse by hand: `math.isqrt(n)` gives the exact integer square root with no float involved; `math.fsum(values)` adds a sequence of floats with far less accumulated

error than the built-in `sum`; `math.comb(n, k)` and `math.perm(n, k)` give combinations and permutations; and `math.prod(values)` multiplies a sequence as `sum` adds one.

THE LAST TRAP: ROUND() ROUNDS TO EVEN

Python's built-in `round` does not round halves up the way most of us learned in school. It uses *banker's rounding*, rounding a value that sits exactly halfway to the nearest *even* digit. So `round(0.5)` is `0`, `round(1.5)` is `2`, and `round(2.5)` is also `2`. This is mandated by IEEE 754 and is statistically less biased than always rounding up, but it astonishes people who do not expect it. When you need school-style rounding, use `Decimal` with an explicit `ROUND_HALF_UP`.

EXAMPLE 4.4 The toolkit in use

PYTHON

```python
import math, statistics

math.isqrt(10**18)             # 1000000000, exact, no float error
math.fsum([0.1] * 10)          # 1.0 exactly; sum([0.1]*10) gives 0.999999999
math.comb(52, 5)               # 2598960, five-card poker hands

statistics.mean([82, 91, 67])   # 80
statistics.stdev([82, 91, 67])  # 12.165...

round(2.5)                     # 2, banker's rounding, not 3
round(3.5)                     # 4
```

→ **FURTHER READING: THE NUMERIC TOOLKIT**

- The math module *Python docs*
- The statistics module *surprisingly capable for everyday stats*
- "How to Round Numbers in Python" *Real Python, on banker's rounding*

§ § §

Exercises

EXERCISE 4.1 Beyond 64 bits

Compute `2 ** 200` and confirm with `.bit_length()` that the result needs 201 bits. Explain in one sentence why this calculation, which would silently overflow a 64-bit integer in C, simply works in Python.

EXERCISE 4.2 The 0.1 problem

Evaluate `0.1 + 0.1 + 0.1 == 0.3` and explain the result. Then rewrite the comparison using `math.isclose` so that it gives the answer a human would expect.

EXERCISE 4.3 Money done right

A shop sells an item for 19.99 with 8.25 percent tax, and a customer buys three. Compute the total two ways: once with ordinary floats, once with `Decimal` built from strings and rounded to the nearest cent. Show that the float version can leave a fraction of a cent, and explain why the `Decimal` version is the one an accountant would accept.

EXERCISE 4.4 Recover the fraction

The float `0.333333333333` is meant to represent one third. Use `Fraction` and `limit_denominator` to recover `Fraction(1, 3)` from it. Then show that `Fraction(1, 3) + Fraction(1, 3) + Fraction(1, 3)` equals exactly `1`, something the float version cannot promise.

EXERCISE 4.5 Count with booleans

Given `temps = [12, 5, 22, 18, 3, 27, 9]`, use the fact that `bool` is an `int` to count, in a single expression with `sum`, how many temperatures are above 15. Do not use a loop or a list comprehension that builds a list.

EXERCISE 4.6 The sign of the remainder

Predict the values of `-7 // 2`, `-7 % 2`, `7 // -2`, and `7 % -2` before running them. Then run them and reconcile any surprises with the rule that the remainder's sign follows the divisor.

EXERCISE 4.7 nan behaves strangely

Create a list `[1.0, float("nan"), 2.0]` and try to check whether `float("nan")` is "in" it using the `in` operator. Then try sorting the list. Explain both results in terms of the rule that `nan` is not equal to itself.

EXERCISE 4.8 A research question

Pick the right numeric type for each of these and justify your choice in one sentence: (a) the number of users registered on a website; (b) the balance of a bank account; (c) the position of a particle in a physics simulation; (d) the exact probability of drawing a specific poker hand; (e) the impedance of an electrical circuit.

Solutions to Chapter 4 Exercises

Solution 4.1: Beyond 64 bits

```python
n = 2 ** 200
print(n.bit_length())   # 201
```

It works because a Python `int` is not a fixed-size machine word but an object holding a growable array of digits, so when a value needs more bits Python allocates a longer array instead of overflowing.

Solution 4.2: The 0.1 problem

`0.1 + 0.1 + 0.1 == 0.3` is `False`, because each `0.1` is a tiny bit off from the true one-tenth, the three errors accumulate, and the sum lands just beside the (also slightly

off) stored value of `0.3`. The correct comparison is `math.isclose(0.1 + 0.1 + 0.1, 0.3)`, which is `True` because it allows for a small tolerance rather than demanding bit-for-bit equality.

Solution 4.3: Money done right

```python
from decimal import Decimal, ROUND_HALF_UP
# float version
print(19.99 * 1.0825 * 3)            # 64.927425000000005
# Decimal version
price, tax = Decimal("19.99"), Decimal("0.0825")
total = (price * (1 + tax) * 3).quantize(Decimal("0.01"), ROUND_HALF_UP)
print(total)                                 # 64.93
```

The float result carries a trailing `...005`, a fraction of a cent that has no meaning in money and that, summed across millions of transactions, becomes real error. The `Decimal` version computes in base ten and rounds explicitly to the cent, so it is exact and auditable, which is what an accountant requires.

Solution 4.4: Recover the fraction

```python
from fractions import Fraction
f = Fraction(0.333333333333).limit_denominator(1000)
print(f)                             # 1/3
print(f + f + f)                     # 1
```

`limit_denominator` finds the simplest fraction within a small tolerance of the float, recovering the one third the number was meant to be. Because `Fraction` stores an exact ratio of integers, adding it to itself three times gives exactly `1`, whereas `0.333333333333 * 3` falls slightly short.

Solution 4.5: Count with booleans

```python
temps = [12, 5, 22, 18, 3, 27, 9]
sum(t > 15 for t in temps)           # 3
```

Each comparison `t > 15` yields `True` or `False`, which are `1` and `0` as integers, so summing the generator counts how many comparisons were true. The values 22, 18, and 27 exceed 15, giving 3.

Solution 4.6: The sign of the remainder

`-7 // 2` is `-4` (floor division rounds down, toward negative infinity, not toward zero). `-7 % 2` is `1` (the remainder's sign follows the positive divisor). `7 // -2` is `-4` (again rounding down). `7 % -2` is `-1` (the divisor is negative, so the remainder is too). The invariant that ties them together is `(a // b) * b + (a % b) == a`, which holds in every case.

Solution 4.7: nan behaves strangely

Sorting raises no error but produces a meaningless order, because the comparisons used by the sort all involve `nan`, which returns `False` for every ordering question, so `nan` lands wherever it happened to start. The membership test is subtler: `float("nan") in [1.0, float("nan"), 2.0]` may return `True`, because `in` first checks identity (`is`) and only then equality, and if it is the very same `nan` object the identity check succeeds even though `nan == nan` is `False`. Both behaviours trace to the IEEE rule that `nan` is unequal to everything, itself included.

Solution 4.8: A research question

(a) `int`: a count is a whole number with no upper bound worth fearing. (b) `Decimal`: money must be exact in base ten and auditable. (c) `float`: physical positions are approximate anyway, and speed matters in a simulation with millions of steps. (d) `Fraction`: a probability like "combinations over total" is exactly rational, and keeping it exact avoids rounding. (e) `complex`: impedance has a real (resistance) and imaginary (reactance) part by its very definition.

CHAPTER 5 · PART II · THE PRIMITIVES

Strings & Text

Most real programs spend most of their time moving text around: reading it, cleaning it, searching it, assembling it into reports. Python's handling of text is among the finest in any language, and it rests on one decision that took the industry fifty years to get right: a string is a sequence of characters, not a sequence of bytes.

> *We were eating dinner in a New Jersey diner. Ken wrote it on the back of a placemat. By the time we got back to the office, UTF-8 was essentially done.*
>
> — Rob Pike, recalling the 1992 design of the UTF-8 encoding with Ken Thompson

There is a quiet revolution buried in the difference between Python 2 and Python 3, and it is almost entirely about text. In the older language, a string was a sequence of bytes, and the unspoken assumption that one byte equalled one character worked beautifully for English and collapsed for everyone else. Python 3 made the harder, correct choice: a string is a sequence of *characters*, abstract symbols drawn from every writing system humanity has ever devised, and bytes are a separate thing you convert to and from deliberately. This single change broke a great deal of old code and saved a generation of new programmers from a category of bug so common it has its own name. Understanding it is the foundation on which everything else in this chapter rests.

We will start with what a string actually is, take a necessary detour through Unicode and encodings (the part most books skip and most programmers later wish they had not), then cover the practical craft: writing string literals, slicing them, the methods that do the daily work, the three generations of formatting, how to build strings without quadratic slowdown, the binary counterpart in `bytes`, and a first proper look at regular expressions.

5.1 What a String Really Is

A Python string is an immutable sequence of **Unicode code points**. A code point is simply a number assigned to a character by the Unicode standard: the capital letter A is code point 65, the Greek alpha is 945, the Devanagari letter क is 2325, and the snake emoji is 128013. A string is a sequence of these numbers, and the characters you see on screen are how they are drawn. Crucially, one element of a Python string is one code point, regardless of how many bytes that code point will take when it is eventually stored or transmitted.

Two built-in functions convert between a character and its code point, and they are the cleanest way to see the principle in action.

```python
ord("A")        # 65  , the code point of a character
chr(65)         # 'A' , the character for a code point
ord("🐍")       # 128013
chr(128013)     # '🐍'

len("🐍")       # 1   , ONE code point, even though it is an emoji
```

That final line is worth dwelling on. In Python 3, the length of the snake emoji is one, because it is one character. In JavaScript and Java, whose strings are sequences of sixteen-bit units, the same emoji has a length of two, because it does not fit in sixteen bits and must be stored as a pair. Python spares you that complication entirely: a character is a character, and its length is one.

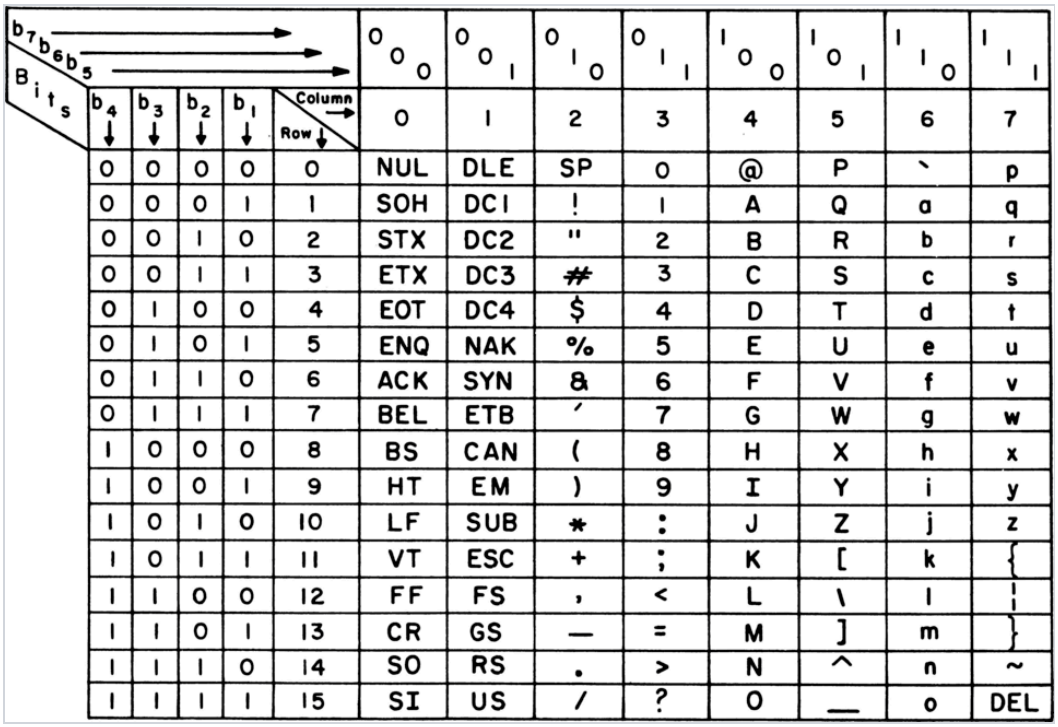
DEF **Code Point**

An integer between 0 and 1,114,111 (0x10FFFF) that the Unicode standard assigns to a single character, written in the form U+0041 for the letter A. A Python string is a sequence of code points. How those code points are turned into actual bytes for storage or transmission is a separate matter, called encoding, and is the subject of the next section.

→ **FURTHER READING: STRINGS AS CODE POINTS**

- Python Unicode HOWTO *the official guide, excellent*
- The Unicode Consortium *the standard itself*

5.2 Unicode and Encodings



Bits b_4	b_3	b_2	b_1	Row ↓ / Column →	0 (000)	1 (001)	2 (010)	3 (011)	4 (100)	5 (101)	6 (110)	7 (111)
0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
0	0	1	0	2	STX	DC2	"	2	B	R	b	r
0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
0	1	0	0	4	EOT	DC4	$	4	D	T	d	t
0	1	0	1	5	ENQ	NAK	%	5	E	U	e	u
0	1	1	0	6	ACK	SYN	&	6	F	V	f	v
0	1	1	1	7	BEL	ETB	'	7	G	W	g	w
1	0	0	0	8	BS	CAN	(	8	H	X	h	x
1	0	0	1	9	HT	EM	)	9	I	Y	i	y
1	0	1	0	10	LF	SUB	*	:	J	Z	j	z
1	0	1	1	11	VT	ESC	+	;	K	[	k	{
1	1	0	0	12	FF	FS	,	<	L	\	l	¦
1	1	0	1	13	CR	GS	—	=	M	]	m	}
1	1	1	0	14	SO	RS	.	>	N	^	n	~
1	1	1	1	15	SI	US	/	?	O	_	o	DEL

Figure 5.1. *The US-ASCII code chart of 1968. With seven bits it could encode 128 symbols, enough for English but for nothing else. The inadequacy of this chart for the rest of the world's writing is the whole reason Unicode had to be invented.* ASCII

For the first decades of computing, the dominant character set was ASCII, which used seven bits to encode 128 symbols: the English alphabet in two cases, the digits, punctuation, and some control codes. It was an American standard for an American industry, and it had no room for the accented letters of French, the umlauts of German, the Cyrillic of Russian, or the thousands of characters of Chinese. A patchwork of incompatible eight-bit and multi-byte extensions grew up to fill the gap, and text written under one scheme turned to garbage when read under another. The garbage was so common that the Japanese gave it a name, *mojibake*, "character transformation".

Unicode, begun in 1987, was the fix: a single universal catalogue assigning a unique code point to every character in every script ever used by humans, with room for more than a million. But Unicode only assigns the numbers. It does not say how to store them as bytes, and that is the job of an *encoding*. The same code point can be turned into bytes in several ways, and the dominant way, by an overwhelming margin, is UTF-8.

> **HISTORY** UTF-8 was designed in September 1992 by Ken Thompson and Rob Pike, the creators of Unix and later Go, over dinner at a diner in New Jersey, with the first draft

written on a placemat. Its genius is that the 128 ASCII characters keep their original single-byte values, so every ASCII file is already a valid UTF-8 file, while other characters expand to two, three, or four bytes. This backward compatibility is why UTF-8 conquered the web: adopting it cost nothing for existing English text.

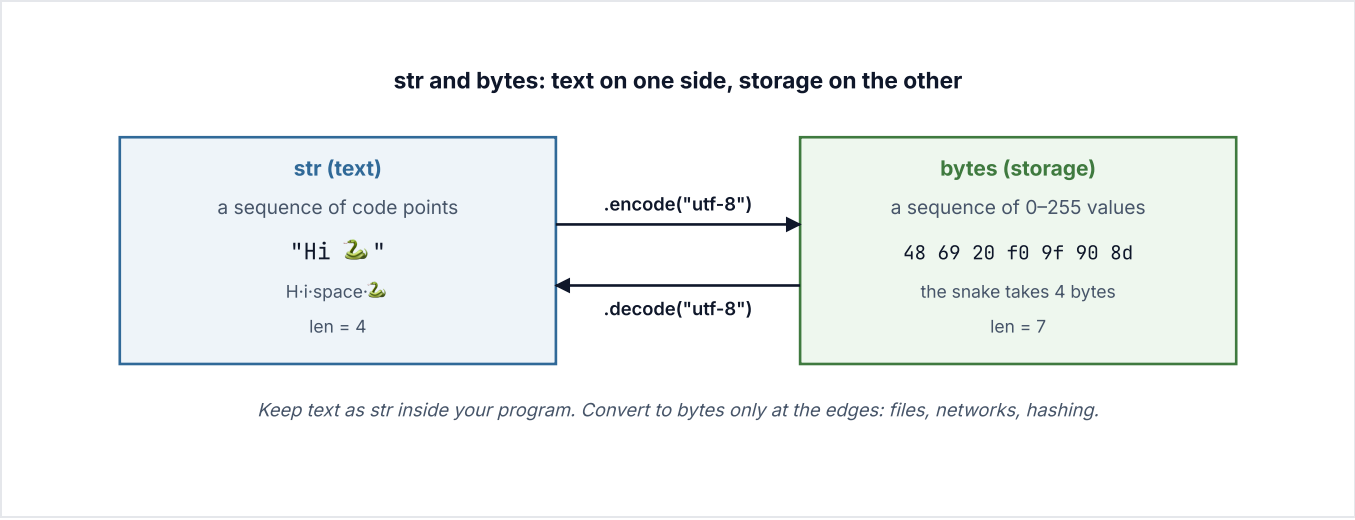


Figure 5.2. The bridge between text and storage. `encode` turns a string into bytes; `decode` turns bytes back into a string. Reading a file or a network message gives you bytes, which you decode; writing means encoding.

EXAMPLE 5.1 Encoding, decoding, and how the bytes differ

PYTHON

```python
s = "café 🐍"
len(s)                          # 6 characters

raw = s.encode("utf-8")         # turn text into bytes
type(raw)                       # <class 'bytes'>
len(raw)                        # 9 bytes: 'é' takes 2, '🐍' takes 4

raw.decode("utf-8")             # 'café 🐍', back to text

# Decoding with the WRONG encoding produces mojibake or an error:
raw.decode("latin-1")           # 'cafÃ© ð\x9f\x90\x8d', garbage
```

The last line is the bug that haunted text processing for decades: the bytes were correct, but they were interpreted under the wrong encoding, and the result is nonsense. The rule that prevents it is simple and absolute: always know the encoding of the bytes you are decoding, and when in doubt, assume UTF-8, because on the modern web it is almost always right.

→ FURTHER READING: UNICODE AND ENCODINGS

- Joel Spolsky: "The Absolute Minimum Every Developer Must Know About Unicode" *the classic essay*
- "The Absolute Minimum Every Software Developer Must Know About Unicode in 2023" *a modern update*
- UTF-8 *Wikipedia, with the encoding scheme*

5.3 How CPython Stores Strings

If a string is a sequence of code points, and code points range up to a million, does every character cost four bytes of memory? For most strings that would be wasteful, since a string of pure English needs only one byte per character. CPython solves this elegantly with a scheme introduced in version 3.3 and described in PEP 393, the *flexible string representation*. The interpreter looks at the largest code point in a string and picks the narrowest storage width that can hold it: one byte per character if every code point fits in 255, two bytes if every code point fits in 65,535, and four bytes only if some character truly needs it.

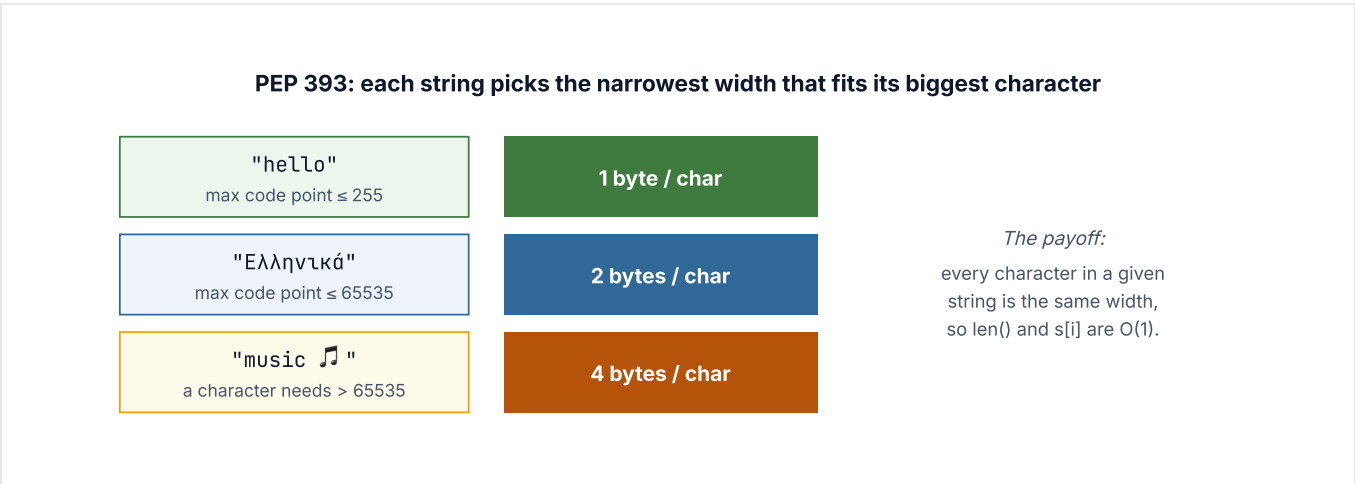


Figure 5.3. The flexible string representation. Because every character within one string occupies the same number of bytes, Python can find the i-th character by simple arithmetic, making indexing and length instant.

This design buys two things at once. It keeps the common case, ASCII and Latin text, as compact as the old byte strings were, so upgrading to proper Unicode cost almost no memory. And because every character in a given string is the same width, the interpreter can jump straight to the i-th character by multiplying, which means indexing a string and asking its length are both instant, no matter how long the string or how exotic its charac-

ters. Languages that store text as UTF-8 internally cannot promise that; they must walk the string to find a character by position. Python's choice trades a little memory for guaranteed fast indexing, and for most programs it is the right trade.

→ **FURTHER READING: STRING STORAGE**

- PEP 393: Flexible String Representation *the design, in full*

5.4 Writing String Literals

Python offers several ways to write a string in source code, each suited to a different need. The choice among them is mostly about avoiding the need to escape characters, which keeps text readable.

PYTHON

```python
a = 'single quotes'
b = "double quotes"            # identical; pick whichever avoids escaping
c = "she said \"hi\""           # escape the inner quotes
d = 'she said "hi"'             # cleaner: use the other quote

e = """A triple-quoted string
spans multiple lines
and keeps the line breaks."""

f = r"C:\Users\Ada\new"          # raw: backslashes are literal, not escapes

g = ("Python automatically joins "   # adjacent string literals
     "two literals with no operator "  # become one string
     "between them.")
```

The raw string, prefixed with `r`, deserves emphasis because it prevents a common class of error. Inside an ordinary string, a backslash begins an escape sequence: `\n` is a newline, `\t` a tab, `\\` a single backslash. This is a problem for Windows file paths and for regular expressions, both of which are full of backslashes. A raw string turns the escaping off, so `r"\n"` is a backslash followed by an `n`, two characters, not one newline. You will use raw strings constantly once you reach regular expressions in Section 5.10.

→ FURTHER READING: STRING LITERALS

○ String and bytes literals *the lexical rules, including every escape sequence*

5.5 Indexing, Slicing, and Immutability

A string is a sequence, so it supports the same indexing and slicing as any sequence, which we will meet again with lists in Chapter 8. Each character has an index from zero, negative indices count from the end, and a slice extracts a range using the notation `[start:stop:step]`, where `stop` is exclusive.

```python
s = "Pythonic"
s[0]        # 'P'
s[-1]       # 'c', last character
s[0:6]      # 'Python', indices 0 through 5
s[:6]       # 'Python', start defaults to 0
s[6:]       # 'ic'    , stop defaults to the end
s[::2]      # 'Pton'  , every second character
s[::-1]     # 'cinohtyP', reversed
```

The fact you must never forget is that strings are **immutable**, exactly as Chapter 3 described. You cannot change a character in place; an attempt raises a `TypeError`. Every operation that appears to modify a string in fact creates a new one. This is not a limitation to resent but a guarantee to rely on: because a string can never change underneath you, you can pass it to any function, use it as a dictionary key, and share it freely, knowing no one can alter it. The one place immutability bites is when you build a long string piece by piece in a loop, and Section 5.8 shows the right way to do that.

5.6 The String Methods

The string type carries roughly forty-seven methods. You do not need to memorise them; you need to know the two dozen that do the daily work and to recognise the shape of the rest so you can look them up. The table below collects the working set.

Method	Does	Example result

`.upper() .lower()`	change case	`"Hi".lower()` → `'hi'`
`.title() .capitalize()`	title / sentence case	`"war".capitalize()` → `'War'`
`.casefold()`	aggressive case fold for matching	`"ß".casefold()` → `'ss'`
`.strip() .lstrip() .rstrip()`	remove surrounding whitespace	`" hi ".strip()` → `'hi'`
`.split(sep)`	break into a list	`"a,b".split(",")` → `['a','b']`
`.join(seq)`	combine a list into a string	`",".join(['a','b'])` → `'a,b'`
`.replace(a, b)`	substitute text	`"cat".replace("c","b")` → `'bat'`
`.find(x) .index(x)`	locate a substring	`"hello".find("ll")` → `2`
`.count(x)`	count occurrences	`"banana".count("a")` → `3`
`.startswith() .endswith()`	test a prefix or suffix	`"a.csv".endswith(".csv")` → `True`
`.isdigit() .isalpha()`	test character classes	`"42".isdigit()` → `True`
`.zfill(n) .center(n)`	pad to a width	`"7".zfill(3)` → `'007'`

Two of these carry more weight than the rest and deserve a closer look, because together they handle a huge fraction of real text work: `split` takes a string apart into a list, and `join` puts a list back together into a string. They are inverses, and the asymmetry in how `join` is called (you call it on the separator, not on the list) trips up every beginner exactly once.

EXAMPLE 5.2 split and join, the workhorses

```python
line = "Ada,Lovelace,1815,London"
fields = line.split(",")
# ['Ada', 'Lovelace', '1815', 'London']

# join is called on the SEPARATOR, with the list as its argument:
" | ".join(fields)
# 'Ada | Lovelace | 1815 | London'

# split() with no argument splits on any run of whitespace, which is
# usually what you want for free-form text:
"  the   quick brown\tfox  ".split()
# ['the', 'quick', 'brown', 'fox']
```

Calling `join` on the separator reads strangely at first but is logical: the separator is the thing that knows how to glue, and it can glue any sequence handed to it. Write it wrong once, see the error, and you will remember it forever.

→ **FURTHER READING: STRING METHODS**

- String methods *the complete reference*

5.7 Formatting: Three Generations

Assembling a string from values, a name, a number, a date, is so common that Python has accumulated three ways to do it over its history. You should write all new code in the third, but you will read the other two in older code and must recognise them.

```python
name, score = "Ada", 91.5

# 1. Percent formatting (oldest, inherited from C). Still seen in logging.
"%s scored %.1f" % (name, score)        # 'Ada scored 91.5'

# 2. str.format (Python 2.6+). Verbose but still common.
"{} scored {:.1f}".format(name, score)   # 'Ada scored 91.5'
```

```
# 3. f-strings (Python 3.6+). The modern default. Use these.
f"{name} scored {score:.1f}"              # 'Ada scored 91.5'
```

The f-string, marked by an `f` before the opening quote, is the clearest of the three because the values appear inline, exactly where they will be in the output. Anything inside the braces is a full Python expression, evaluated on the spot, and after a colon you may add a *format specification* controlling width, alignment, precision, sign, and grouping. The specification is its own compact mini-language, and the most useful pieces of it fit in one example.

EXAMPLE 5.3 The f-string format specification

PYTHON

```
n = 1234567
pi = 3.14159265
name = "Ada"

f"{pi:.2f}"          # '3.14'      , two decimal places
f"{n:,}"             # '1,234,567' , thousands separators
f"{n:_}"             # '1_234_567' , underscore grouping
f"{0.875:.1%}"       # '87.5%'     , as a percentage
f"{255:#x}"          # '0xff'      , hexadecimal with prefix
f"{name:>10}"        # '       Ada', right-align in width 10
f"{name:^10}"        # '   Ada    ', centre
f"{name:*<10}"       # 'Ada*******', left-align, pad with *

# The self-documenting form (Python 3.8+), invaluable for debugging:
f"{score=}"          # 'score=91.5', prints the name and the value
```

The self-documenting form, an equals sign just before the closing brace, prints both the expression and its value. It is the fastest way to scatter informative debugging output through a program, and it is worth committing to memory now.

→ **FURTHER READING: FORMATTING**

- Formatted string literals *the f-string reference*
- fstring.help *a cheat sheet for the format spec*
- pyformat.info *format() and % side by side*

5.8 Building Strings Without Quadratic Slowdown

Because strings are immutable, a deceptively innocent loop can become catastrophically slow, and this is one of the few performance traps worth learning before you ever hit it. Consider building one long string by repeatedly adding to it with `+=`. Each `+=` cannot change the existing string, since strings are immutable, so it must create a *new* string by copying every character built so far and then appending the next piece. Building a string of n pieces this way copies roughly n-squared characters in total, which crawls to a halt as n grows.

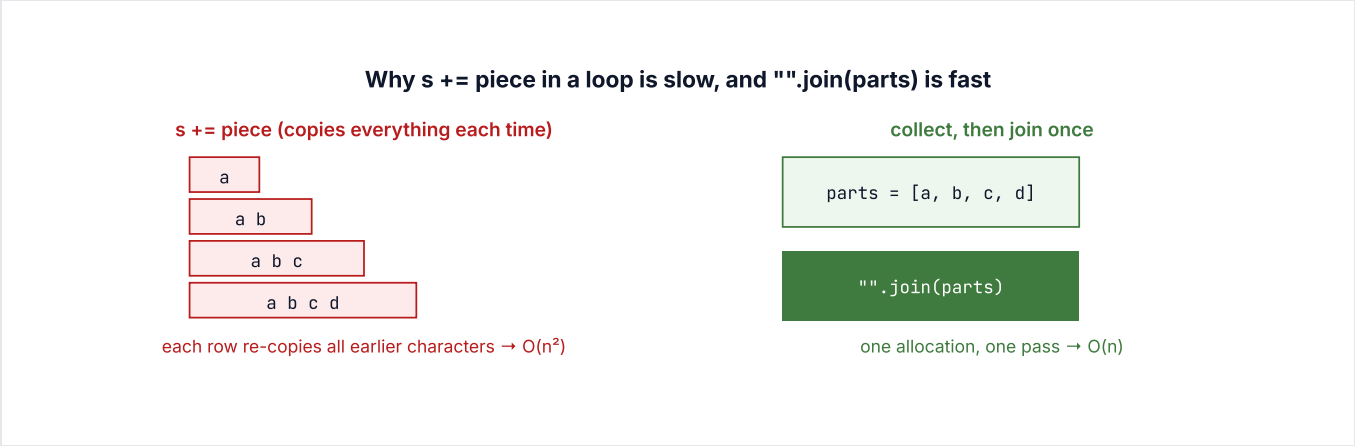


Figure 5.4. The string-building trap. Appending to an immutable string in a loop repeatedly copies the whole accumulated result. Collecting the pieces in a list and joining once at the end does the work in a single pass.

The idiomatic remedy is to collect the pieces in a list, which is mutable and appends cheaply, and to join them in a single operation at the end. The `join` method computes the total length once, allocates the result once, and copies each piece once, turning quadratic work into linear.

PYTHON

```
# Slow: O(n²), re-copies on every iteration
result = ""
for piece in pieces:
    result += piece

# Fast and idiomatic: O(n)
result = "".join(pieces)

# When the pieces are produced incrementally, io.StringIO is the tool:
import io
buffer = io.StringIO()
for piece in pieces:
```

```
    buffer.write(piece)
result = buffer.getvalue()
```

AN HONEST CAVEAT CPython contains a special optimisation that sometimes makes `s += piece` run in linear time when `s` has no other references. It is real but fragile: it does not apply on other implementations like PyPy, and it breaks the moment a second name points at the string. Do not rely on it. The `join` idiom is always fast and always clear, so simply make it a habit.

→ **FURTHER READING: BUILDING STRINGS**

- io.StringIO *Python docs*
- "Efficient String Concatenation in Python" *benchmarks of the approaches*

5.9 Bytes and Bytearray

Text is what your program reasons about; bytes are what crosses its boundaries. When you read a file, receive a network message, compute a cryptographic hash, or talk to hardware, you are handling raw bytes, and Python gives you the `bytes` type for exactly this. A `bytes` object is an immutable sequence of integers from 0 to 255, written with a `b` prefix, and its mutable sibling is `bytearray`.

PYTHON

```python
data = b"hello"
type(data)             # <class 'bytes'>
data[0]               # 104, an integer, not a character
data.decode("utf-8")   # 'hello', turn bytes back into text

# Hashing operates on bytes, not text, so you must encode first:
import hashlib
hashlib.sha256("password".encode("utf-8")).hexdigest()
# '5e884898da28047151d0e56f8dc6292773603d0d6aabbdd62a11ef721d1542d8'
```

The discipline that keeps text handling sane is to convert at the boundaries and reason in text within. Decode bytes to `str` as soon as they enter your program, do all your work on strings, and encode back to bytes only when you write out. This is sometimes called

the "Unicode sandwich": bytes on the outside, text in the middle. Programs that mix the two carelessly are exactly the programs that produce mojibake.

→ FURTHER READING: BYTES

- Binary sequence types *bytes, bytearray, memoryview*
- Ned Batchelder: "Pragmatic Unicode" *the Unicode sandwich, explained*

5.10 A First Look at Regular Expressions

When the text you are searching has a *shape* rather than a fixed form, a phone number, an email address, a date, a log line, plain string methods are not enough, and you reach for the `re` module and its regular expressions. A regular expression is a small, dense pattern language: `\d` matches any digit, `\w` any word character, `.` any character at all, while `*`, `+`, and `?` control repetition. Patterns are almost always written as raw strings, because they are full of backslashes.

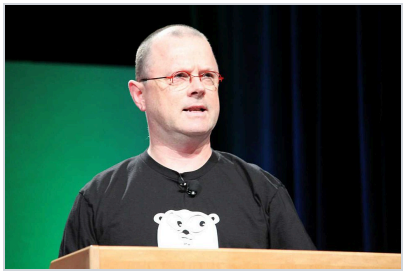


Figure 5.5. *Rob Pike, co-designer of UTF-8, photographed at a conference.*

EXAMPLE 5.4 The four operations you will use most

PYTHON

```python
import re
text = "Reach me at ada@example.com or bob@work.org by Friday."

# findall: every match, as a list
re.findall(r"[\w.]+@[\w.]+", text)
# ['ada@example.com', 'bob@work.org']

# search: the first match anywhere, or None
m = re.search(r"(\w+)@(\w+)", text)
m.group(0)        # 'ada@example', the whole match
m.group(1)        # 'ada'       , first parenthesised group

# sub: replace every match
re.sub(r"[\w.]+@[\w.]+", "[email hidden]", text)
# 'Reach me at [email hidden] or [email hidden] by Friday.'

# fullmatch: does the WHOLE string fit the pattern? (validation)
bool(re.fullmatch(r"\d{4}-\d{2}-\d{2}", "2026-05-27"))   # True
```

USE THEM SPARINGLY

Regular expressions are powerful and easy to abuse. If you are searching for fixed text, use `str` methods like `in`, `find`, or `replace`, which are clearer and faster. If your data is structured, such as JSON, HTML, or CSV, use a real parser, never a regular expression, because those formats are not regular and a regex will eventually meet input it mishandles. Reach for regular expressions when the shape is genuinely regular but variable: identifiers, phone numbers, log formats. A famous saying warns: faced with a problem, some people think "I will use regular expressions", and now they have two problems.

→ **FURTHER READING: REGULAR EXPRESSIONS**

- Python Regular Expression HOWTO *the official tutorial*
- regex101.com *an interactive tester that explains each part*
- regular-expressions.info *the deepest reference on the web*

§ § §

Exercises

EXERCISE 5.1 Characters versus bytes

For the string `s = "naïve 🦦"`, predict `len(s)` and `len(s.encode("utf-8"))` before running them. Explain why the two numbers differ, referring to the idea of code points and encoding.

EXERCISE 5.2 Round trip

Take the string `"Zürich"`, encode it to UTF-8 bytes, then decode it back. Then deliberately decode the same bytes with `"latin-1"` and observe the mojibake. Explain in one sentence what went wrong in the second case.

EXERCISE 5.3 Parse a record

Given the line `"Lovelace, Ada, 1815"`, use `split` and `strip` to produce the list `['Lovelace', 'Ada', '1815']` with no surrounding spaces on any field. Then reassemble it into `"Ada Lovelace (1815)"` using an f-string.

EXERCISE 5.4 The join puzzle

A beginner writes `["a", "b", "c"].join("-")` and gets an `AttributeError`. Explain why, and write the correct version. Why did Python's designers attach `join` to the string rather than to the list?

EXERCISE 5.5 Format a table

Given `rows = [("Ada", 91.5), ("Bo", 88.25), ("Caroline", 100.0)]`, print each row so the names are left-aligned in a width of 10 and the scores are right-aligned with one decimal place in a width of 6. Use f-strings with format specifications.

EXERCISE 5.6 Measure the slowdown

Using the `time` module, build a string of 100,000 characters two ways: with `+=` in a loop, and with `"".join(...)`. Compare the times. Explain the difference in terms of how many characters each approach copies in total.

EXERCISE 5.7 A palindrome, properly

Write a function that decides whether a string is a palindrome, ignoring case, spaces, and punctuation, so that `"A man, a plan, a canal: Panama"` returns `True`. Use string methods and slicing, not a loop. Which method removes the non-letters most cleanly?

EXERCISE 5.8 Regex versus method

You need to check whether a string ends in `.csv`, and separately to extract all numbers from a sentence. One of these should use a string method and one should use a regular expression. Decide which is which, write both, and justify your choice using the guidance in Section 5.10.

Solutions to Chapter 5 Exercises

Solution 5.1: Characters versus bytes

`len(s)` is 7 (n, a, ï, v, e, space, 🎲: seven code points). `len(s.encode("utf-8"))` is 11. Counting the bytes: n, a, v, e, and the space take one each (5 bytes), `ï` takes two, and the die emoji takes four, giving 5 + 2 + 4 = 11. The numbers differ because `len` on a string counts characters (code points), while `len` on the encoded bytes counts the storage those characters require, and non-ASCII characters require more than one byte each.

Solution 5.2: Round trip

```python
raw = "Zürich".encode("utf-8")
raw.decode("utf-8")     # 'Zürich', correct
```

```
raw.decode("latin-1")   # 'ZÃ¼rich', mojibake
```

The `ü` is encoded by UTF-8 as two bytes. Latin-1 interprets each byte as its own character, so those two bytes become two separate, wrong characters (`Ã¼`). The bytes were never corrupted; they were decoded under an encoding different from the one used to create them.

Solution 5.3: Parse a record

```python
line = "Lovelace, Ada, 1815"
parts = [p.strip() for p in line.split(",")]
# ['Lovelace', 'Ada', '1815']
last, first, year = parts
f"{first} {last} ({year})"   # 'Ada Lovelace (1815)'
```

Splitting on the comma leaves a leading space on the second and third fields, so each piece is passed through `strip` to remove it.

Solution 5.4: The join puzzle

Lists have no `join` method, so `["a","b","c"].join("-")` raises `AttributeError`. The correct form is `"-".join(["a", "b", "c"])`, which gives `'a-b-c'`. `join` belongs to the string because the separator is the thing doing the joining and because `join` can then glue *any* iterable of strings, not only lists; attaching it to the separator made it general.

Solution 5.5: Format a table

```python
rows = [("Ada", 91.5), ("Bo", 88.25), ("Caroline", 100.0)]
for name, score in rows:
    print(f"{name:<10}{score:>6.1f}")
```

```
Ada         91.5
Bo          88.2
Caroline   100.0
```

`{name:<10}` left-aligns the name in ten columns; `{score:>6.1f}` right-aligns the score in six columns with one decimal place. Note that 88.25 displays as 88.2 because of banker's rounding from Chapter 4.

Solution 5.6: Measure the slowdown

```python
import time
n = 100_000

t = time.perf_counter()
s = ""
for _ in range(n): s += "x"
print("+=", time.perf_counter() - t)

t = time.perf_counter()
s = "".join("x" for _ in range(n))
print("join", time.perf_counter() - t)
```

The `join` version is dramatically faster (often by a factor of hundreds, though CPython's special-case optimisation can narrow the gap on this exact pattern). In principle, `+=` copies all characters accumulated so far on every one of the n iterations, doing work proportional to n-squared, while `join` allocates the result once and copies each character once, doing work proportional to n.

Solution 5.7: A palindrome, properly

```python
def is_palindrome(s):
    cleaned = "".join(c.lower() for c in s if c.isalnum())
    return cleaned == cleaned[::-1]

is_palindrome("A man, a plan, a canal: Panama")   # True
```

The comprehension keeps only alphanumeric characters (using `isalnum`) and lowercases them, joining the survivors into a clean string. The palindrome test is then the elegant `cleaned == cleaned[::-1]`, comparing the string to its reverse via slicing.

Solution 5.8: Regex versus method

```python
# Fixed suffix: a string method is clearer and faster.
"data.csv".endswith(".csv")        # True

# Variable shape (numbers anywhere): a regular expression fits.
import re
re.findall(r"\d+", "order 17 shipped 2026, qty 3")   # ['17', '2026', '3']
```

Checking for a fixed ending is exactly what `endswith` is for, so a regular expression there would be needless complexity. Extracting numbers that may appear anywhere and have varying length is a genuine pattern-matching task, which is precisely where regular expressions earn their keep.